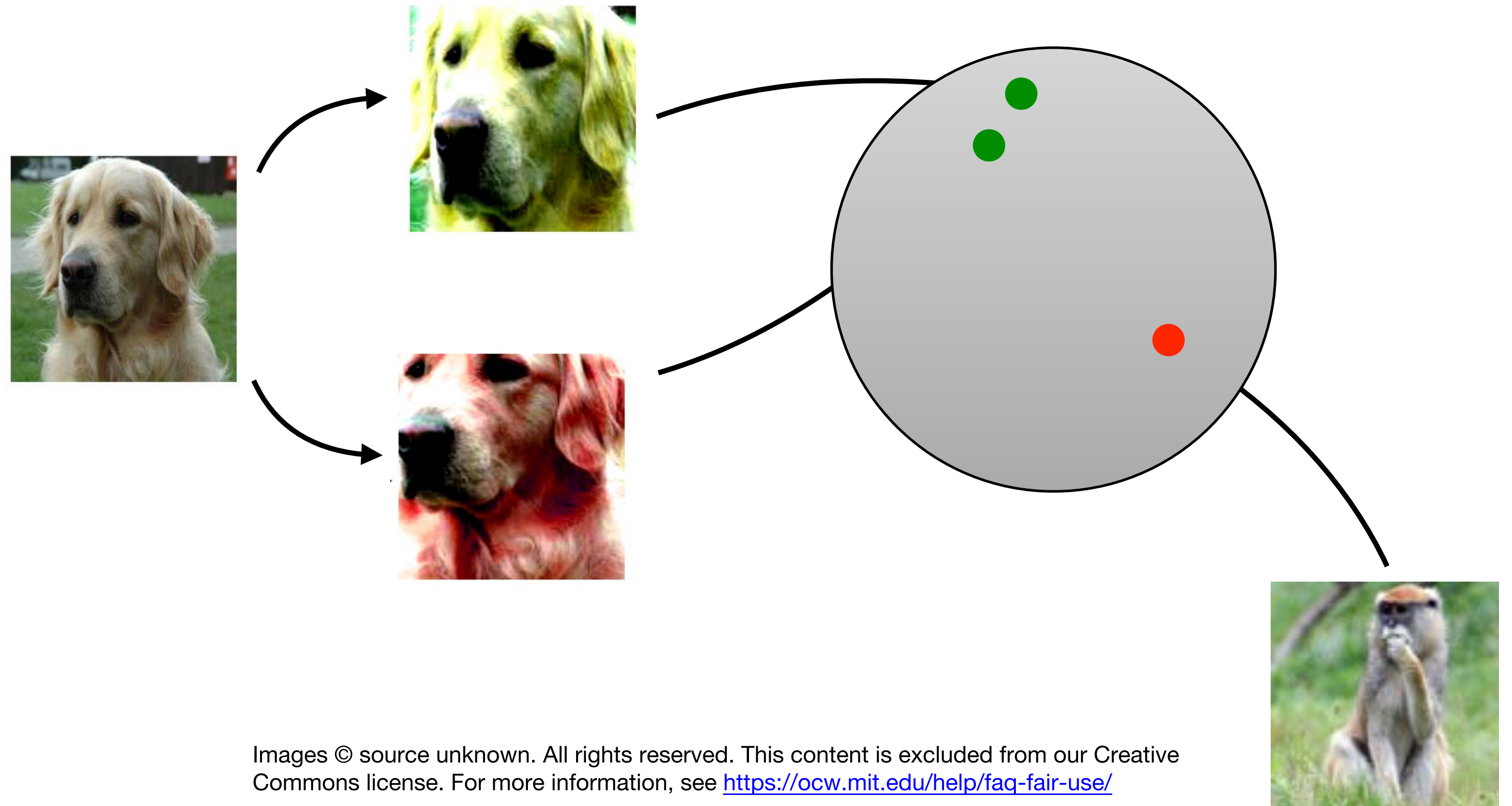


Lecture 12: Similarity-based Representation Learning

Speaker: Sara Beery



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Roadmap: similarity-based representation learning

- Representation learning — why?
- What is a “good” representation?
- Metric learning
- Contrastive representation learning (self-supervised)
 - What does it do?
 - Models

Why learn representations?

- To improve generalization
- To do more learning (transfer learning)
- To exploit geometric similarity for new data or queries:
 - Have we seen the face of this person before or is it new?
 - Retrieval: which items are similar to the query?
- To improve clustering with side information (similar/dissimilar pairs)
- Dimensionality reduction (often unsupervised)

What do we expect from such representations?

What is a “good” representation?

“Generally speaking, a good representation is one that makes a subsequent learning task easier.” — *Deep Learning*, Goodfellow et al. 2016

What could this mean?

What is a “good” representation?

1. Compact (*minimal*)
2. Explanatory (*sufficient*)

What is a “good” representation?

NeurIPS 2020 Competition:
Predicting Generalization in Deep Learning (**Version 1.1**)

Yiding Jiang ^{*†} Pierre Foret[†] Scott Yak[†] Daniel M. Roy^{‡§}
Hossein Mobahi^{†§} Gintare Karolina Dziugaite^{¶§} Samy Bengio^{†§}
Suriya Gunasekar^{||§} Isabelle Guyon ^{*§} Behnam Neyshabur^{†§}

pgdl.neurips@gmail.com

December 16, 2020

3 winning strategies look at:

- Geometry of representation: consistency, separation
- Robustness to perturbations

What helps generalization?

- Representations of CIFAR-10 data with true and random labels

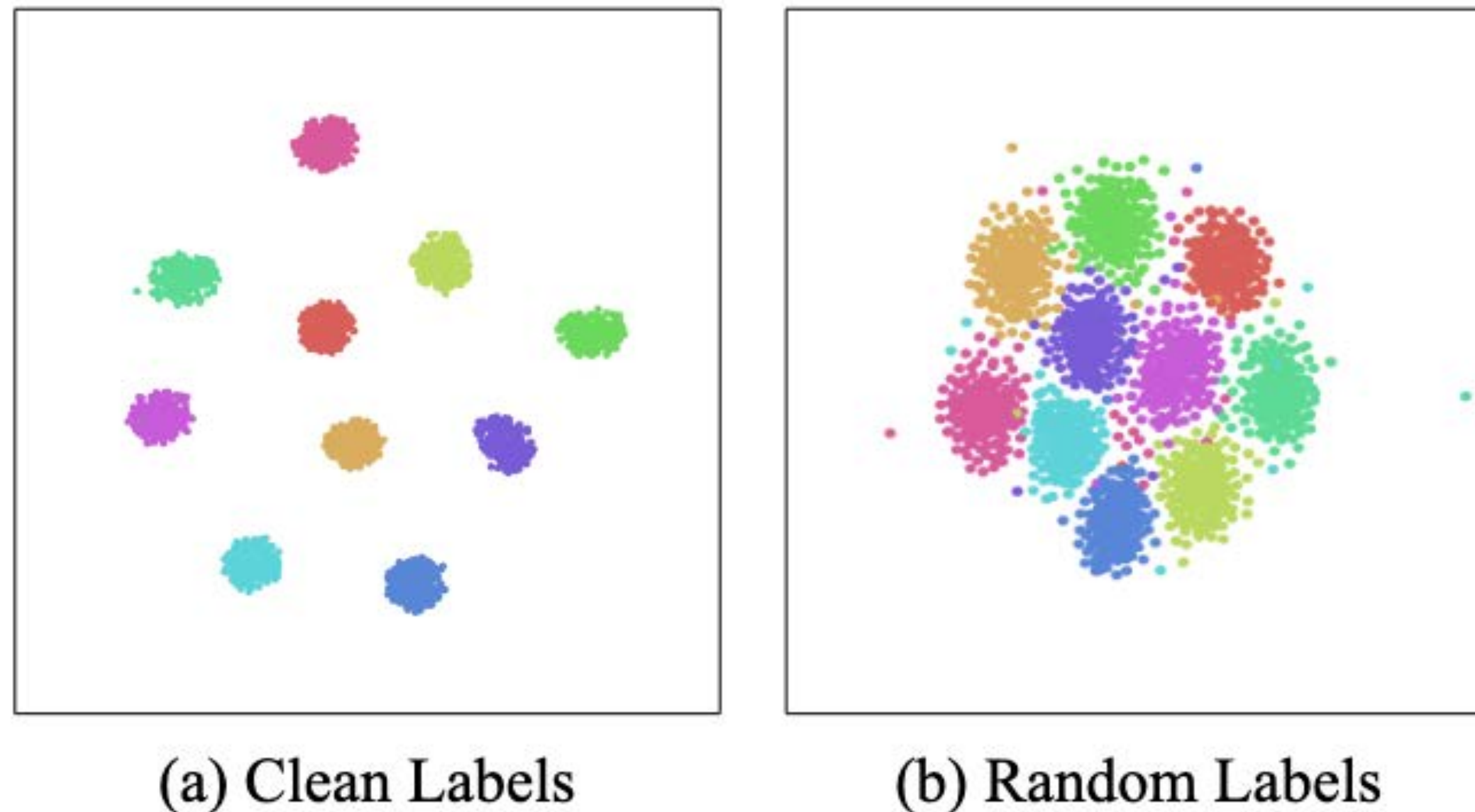


Figure 4: t-SNE visualization of representations.
Classes are indicated by colors.

Courtesy of Chuang, et al. Used under CC BY.

image: Chuang et al., Measuring generalization with optimal transport, 2021

What helps generalization?

- Representations of CIFAR-10 data with true and random labels

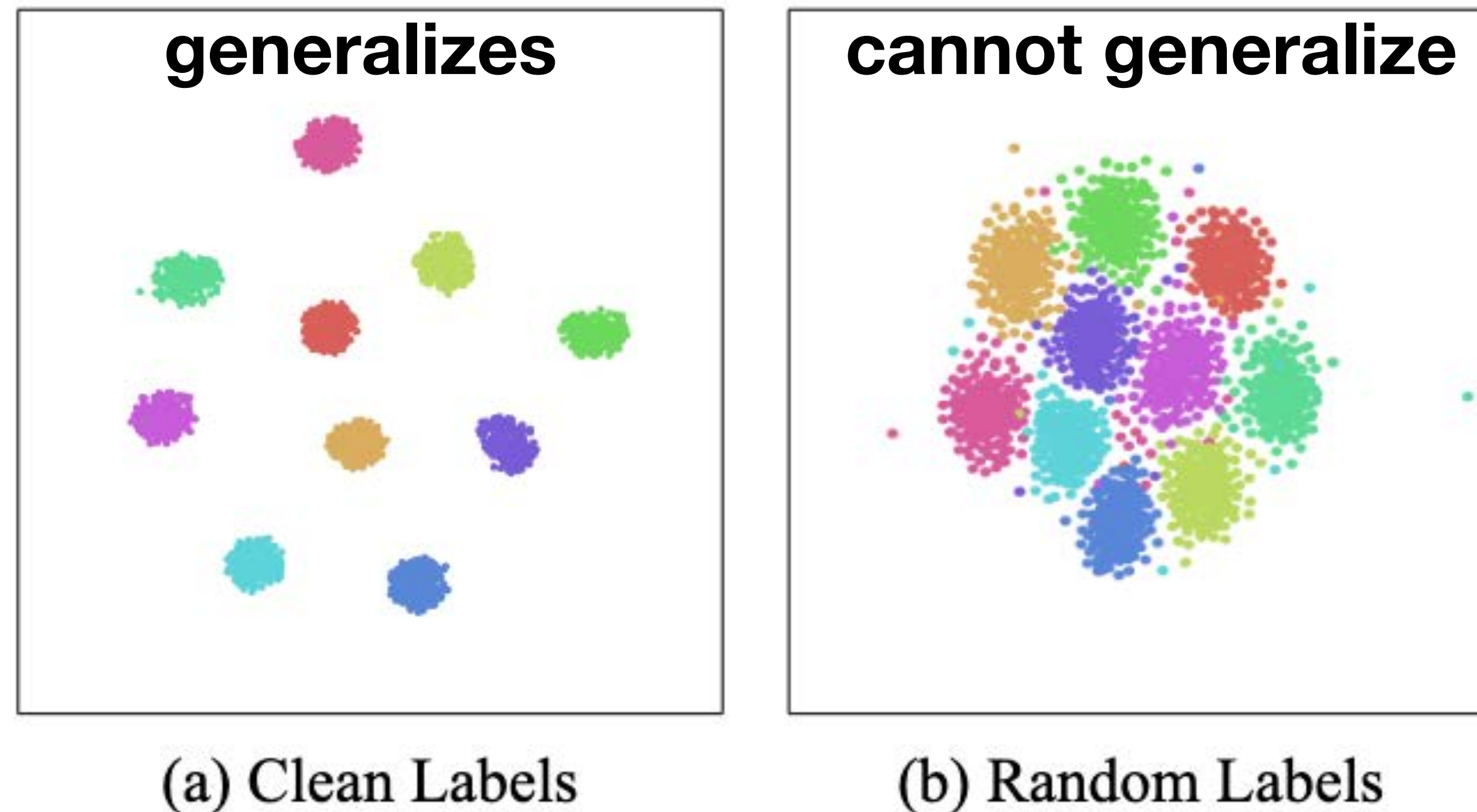


Figure 4: t-SNE visualization of representations. Classes are indicated by colors.

Concentration/consistency:
Data from the same class is close together
Separation: classes are well separated
Robustness

What is a “good” representation?

1. Compact (*minimal*)
2. Explanatory (*sufficient*)
3. Concentration: Data from the same class is close together
4. Separation: classes are well separated
5. Robustness to irrelevant perturbations

How could we encourage a model during training to achieve this?

Similarity-based representation learning

- Encourage good representations via feedback in terms of similarity: pairs of similar/dissimilar inputs

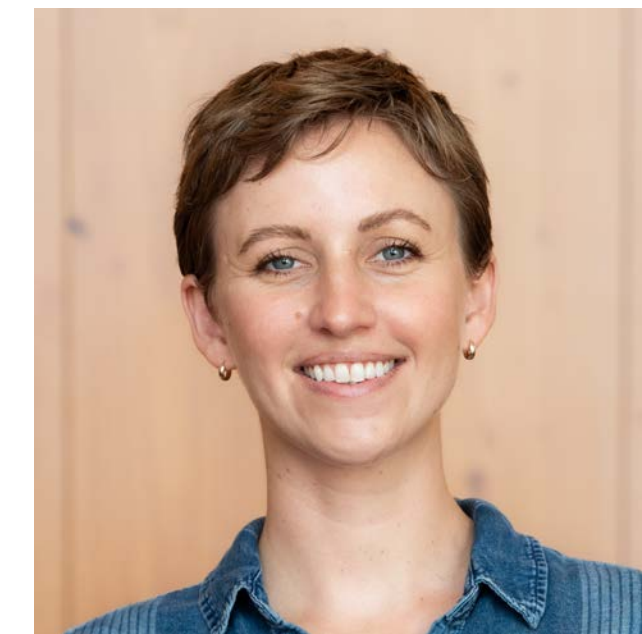


Unsupervised

Supervised

Metric Learning

- Euclidean distance in input space may be not ideal
- Instead: learn a metric that respects desired properties
- Goal: learn a metric where:
 - data points that “belong together” are *similar* (close together)
 - data points that are “different” are *dissimilar* (far apart)
- “Supervision”: similarity information.



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DIAGONALISATION OF THE COVARIANCE MATRIX:

$$\Sigma \rightarrow d \times d.$$

$$\rightarrow \Sigma = \Sigma^T \rightarrow \text{SYMMETRIC.}$$

$\rightarrow \Sigma$ is diagonalizable to a diagonal matrix D via a transformation, $D = U \Sigma U^T$, where U is an orthogonal matrix ($A A^T = A^T A = I \rightarrow \text{ORTHOGONAL}$).



$$\boxed{\Sigma = U^T D U.}$$

$\rightarrow \Sigma$ is +ve semi-definite. , Diagonal entries ≥ 0 .

$\rightarrow \Sigma$ has a unit square-root. $\Sigma^{1/2} \cdot \Sigma^{1/2} = \Sigma$.



$\Sigma^{1/2} \rightarrow \text{unique.}$

$\rightarrow \Sigma \rightarrow d \times d \rightarrow d$ orthonormal eigenvectors
(even if repeated eigen values)

If U is a matrix with rows corresponding to the orthonormal eigen vectors, then the diagonal matrix $D = U \Sigma U^T$ contains the eigen-values of Σ along its diagonal.

Metric learning (linear)

- Data points $\mathbf{x}_1, \dots, \mathbf{x}_n$
- Weak supervision: $\mathcal{S} := \{(\mathbf{x}_i, \mathbf{x}_j) \mid \mathbf{x}_i \text{ and } \mathbf{x}_j \text{ are in the same class}\}$ *similar*
 $\mathcal{D} := \{(\mathbf{x}_i, \mathbf{x}_j) \mid \mathbf{x}_i \text{ and } \mathbf{x}_j \text{ are in different classes}\}$ *dissimilar*
- Goal: learn a linear transformation $\mathbf{z} = \mathbf{W}\mathbf{x}$ that respects similarity
- Use Euclidean distance in representation space:

$$\|\mathbf{z}_i - \mathbf{z}_j\|^2 = (\mathbf{x}_i - \mathbf{x}_j)^\top \mathbf{W}^\top \mathbf{W} (\mathbf{x}_i - \mathbf{x}_j) \quad \mathbf{A} = \mathbf{W}^\top \mathbf{W}$$

Mahalanobis distance with positive semidefinite matrix $\mathbf{A}$, $d_{\mathbf{A}}(\mathbf{x}_i, \mathbf{x}_j) = \sqrt{(\mathbf{x}_i - \mathbf{x}_j)^\top \mathbf{A} (\mathbf{x}_i - \mathbf{x}_j)}$

How can we phrase this as an optimization problem?

“Losses”: upper/lower bound constraints

- first approach (Xing et al 2003):

$$\begin{aligned} \min_{A \succeq 0} \quad & \sum_{(i,j) \in S} d_A(x_i, x_j)^2 && \text{min distance of similar points} \\ \text{s.t.} \quad & \sum_{(k,\ell) \in D} d_A(x_k, x_\ell)^2 \geq 1 && \text{keep distance of dissimilar points} \end{aligned}$$

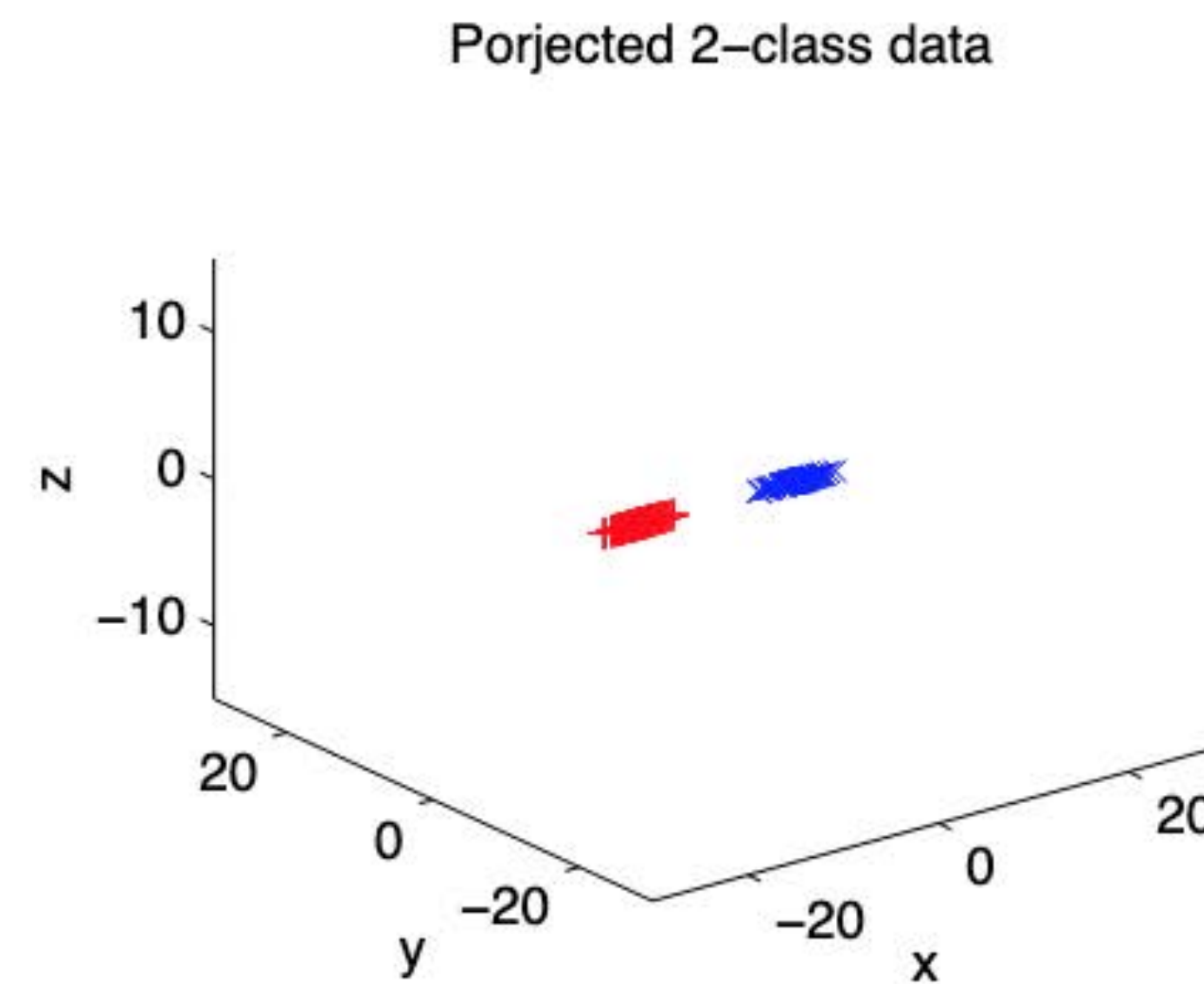
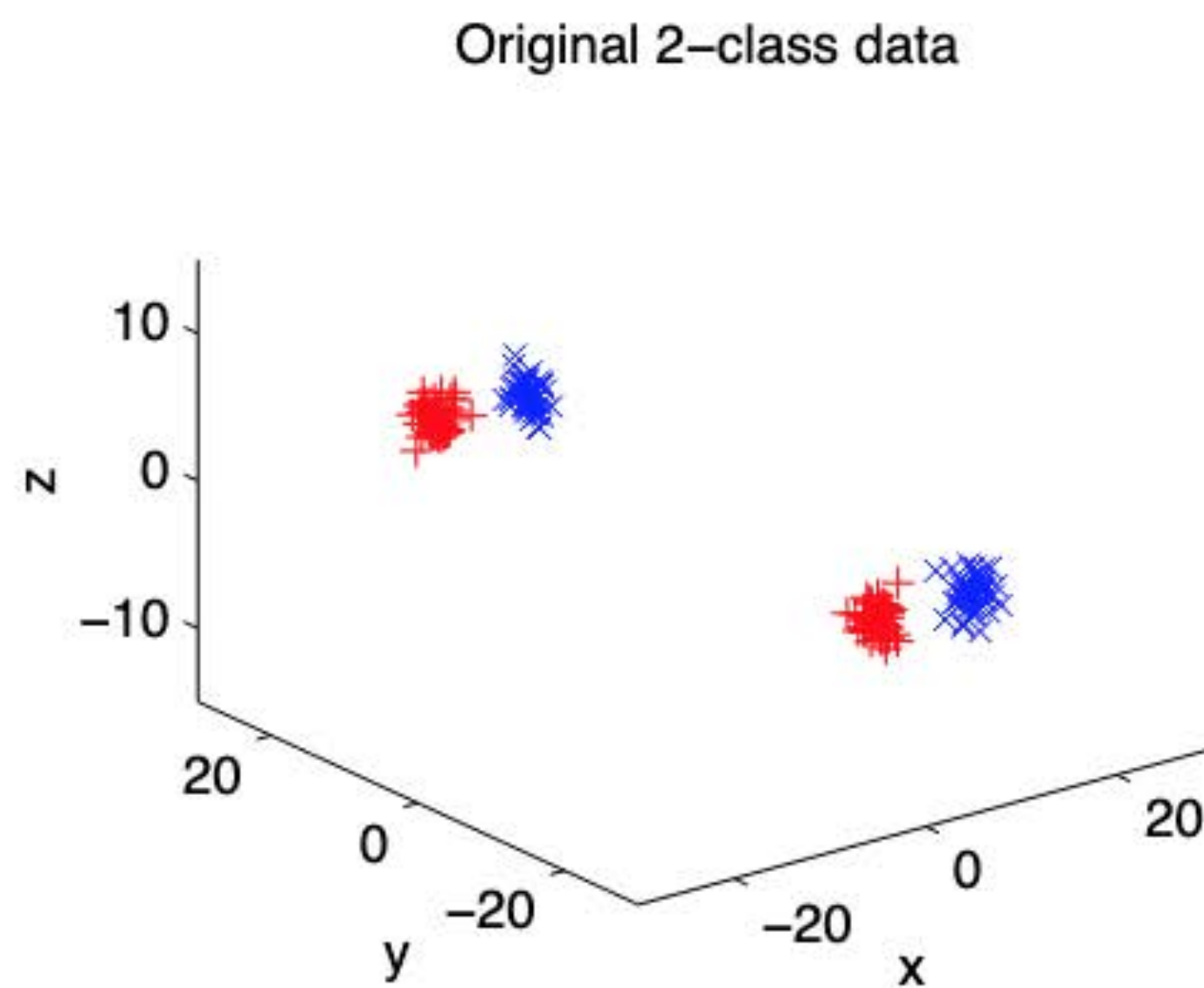
**Distance metric learning, with application
to clustering with side-information**

Eric P. Xing, Andrew Y. Ng, Michael I. Jordan and Stuart Russell
University of California, Berkeley
Berkeley, CA 94720
{epxing, ang, jordan, russell}@cs.berkeley.edu

introduced the term and problem in 2003

- can swap objective and constraint (upper bound for similar pairs)
- many related ideas & follow-ups, e.g.
information-theoretic metric learning (Davis et al 2007):
preserve distribution information (relative entropy between Gaussians) while
observing upper/lower bounds as constraints

Simple example



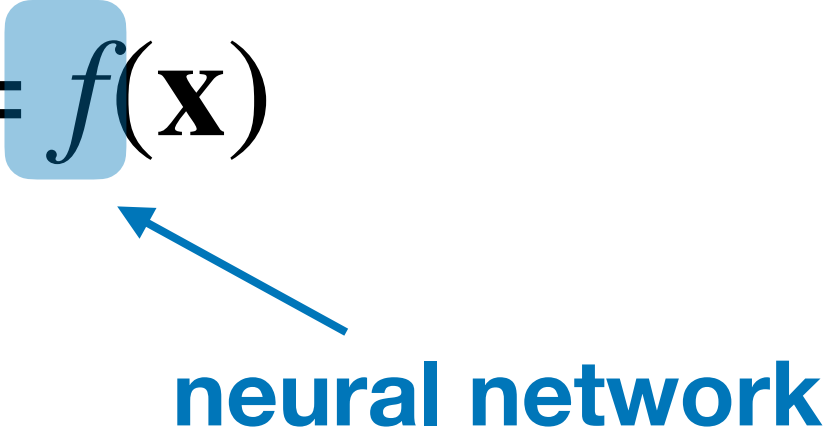
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Improvements / developments

- Nonlinear transformations (kernels, deep metric learning)
- Contrastive losses
- Normalization of representations: angle instead of distance

Deep metric learning

- Linear metric learning: learn a linear transformation $\mathbf{z} = \mathbf{W}\mathbf{x}$

- Deep metric learning: learn a nonlinear transformation $\mathbf{z} = f(\mathbf{x})$ 

optimize not over psd matrices but weights of a neural network

Contrastive losses: intuition



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Contrastive losses

distance of dissimilar pair(s) **distance of similar pair(s)**

- Triplet loss (*Schroff et al 2015*):

$$\mathcal{L}_{\text{triplet}}(\mathbf{x}, \mathbf{x}^+, \mathbf{x}^-) = \sum_{\mathbf{x} \in \mathcal{X}} \max \left(0, \underbrace{\|f(\mathbf{x}) - f(\mathbf{x}^+)\|_2^2}_{\text{distance of similar pair(s)}} - \underbrace{\|f(\mathbf{x}) - f(\mathbf{x}^-)\|_2^2}_{\text{distance of dissimilar pair(s)}} + \epsilon \right)$$

margin

related: Large-margin Nearest Neighbor metric learning (LMNN) (*Weinberger et al 2009*)

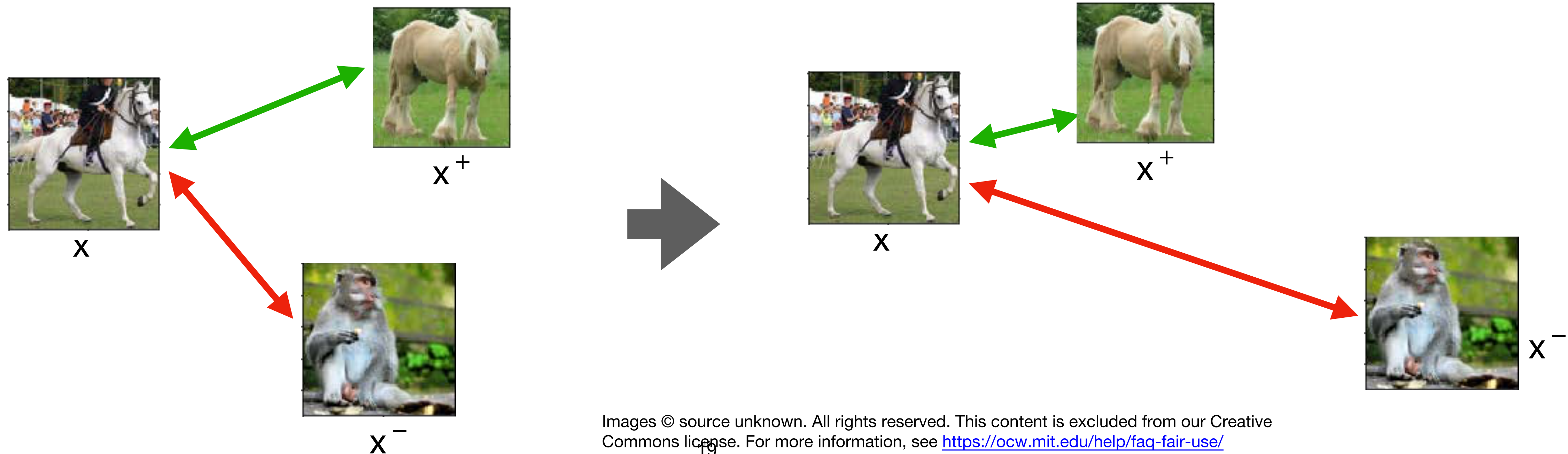
Contrastive losses

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- Triplet loss (Schroff et al 2015):

$$\mathcal{L}_{\text{triplet}}(\mathbf{x}, \mathbf{x}^+, \mathbf{x}^-) = \sum_{\mathbf{x} \in \mathcal{X}} \max \left(0, \underbrace{\|f(\mathbf{x}) - f(\mathbf{x}^+)\|_2^2}_{\text{distance of similar pair(s)}} - \underbrace{\|f(\mathbf{x}) - f(\mathbf{x}^-)\|_2^2}_{\text{distance of dissimilar pair(s)}} + \epsilon \right)$$

margin



Triplet network

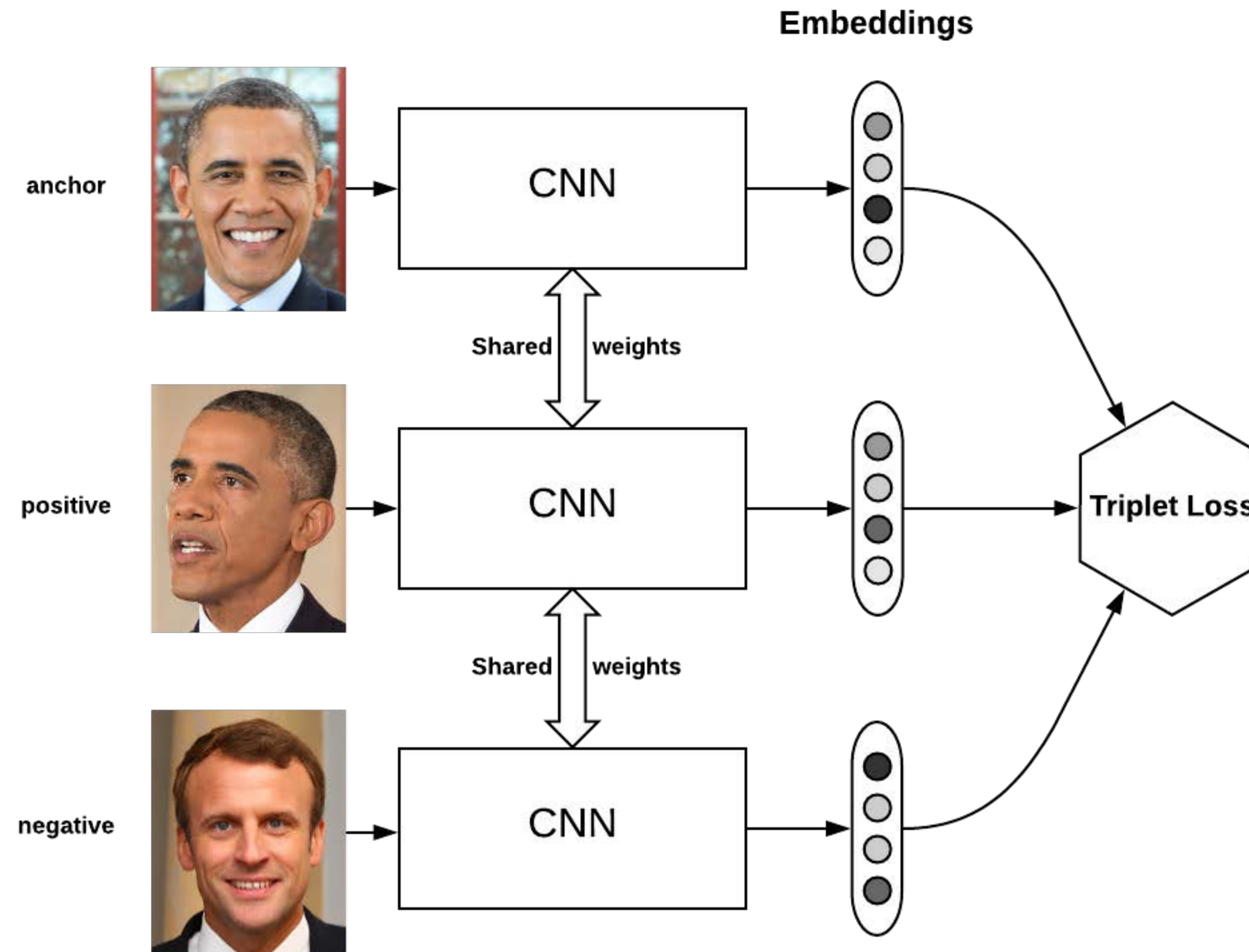


Image © Olivier Moindrot. All rights reserved. This content is excluded from our Creative Commons license. For more information, see <https://ocw.mit.edu/help/faq-fair-use/>

figure: <https://omoindrot.github.io/triplet-loss>

Contrastive losses

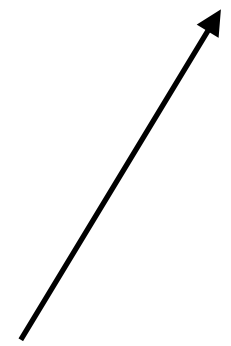
distance of dissimilar pair(s) **distance of similar pair(s)**

- Improvements: compare to multiple negatives per positive pair, e.g.
Lifted structured loss (Song et al 2015): compare to all negatives in a batch

$$\mathcal{L}_{\text{struct}} = \frac{1}{2|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} \max(0, \mathcal{L}_{\text{struct}}^{(ij)})^2$$

where $\mathcal{L}_{\text{struct}}^{(ij)} = D_{ij} + \max \left(\max_{(i,k) \in \mathcal{N}} \epsilon - D_{ik}, \max_{(j,l) \in \mathcal{N}} \epsilon - D_{jl} \right)$

$\|f(x_i) - f(x_j)\|_2$



or smooth relaxation of the max

Contrastive losses

distance of dissimilar pair(s) $\gg$ distance of similar pair(s)

- Triplet loss (Schroff et al 2015):

$$\mathcal{L}_{\text{triplet}}(\mathbf{x}, \mathbf{x}^+, \mathbf{x}^-) = \sum_{\mathbf{x} \in \mathcal{X}} \max \left(0, \underbrace{\|f(\mathbf{x}) - f(\mathbf{x}^+)\|_2^2}_{\text{green}} - \underbrace{\|f(\mathbf{x}) - f(\mathbf{x}^-)\|_2^2}_{\text{red}} + \epsilon \right)$$

↖ margin

Demands:

dist(anchor, negative ex) $>$ dist(anchor, positive ex) + margin

And so what this actually
looks like in practice

related: Large-margin Nearest Neighbor metric
learning (LMNN) (Weinberger et al 2009)

Example embedding

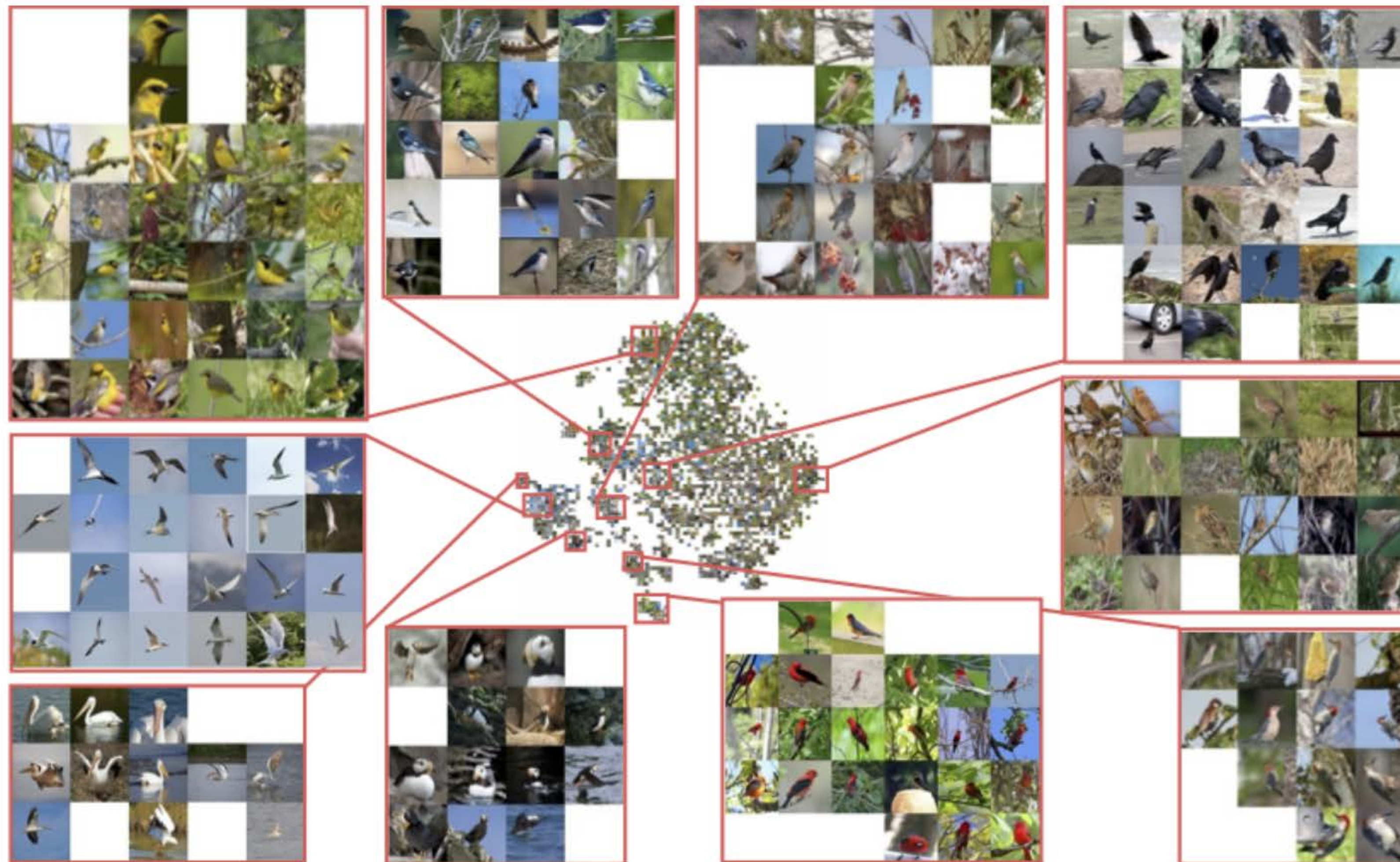
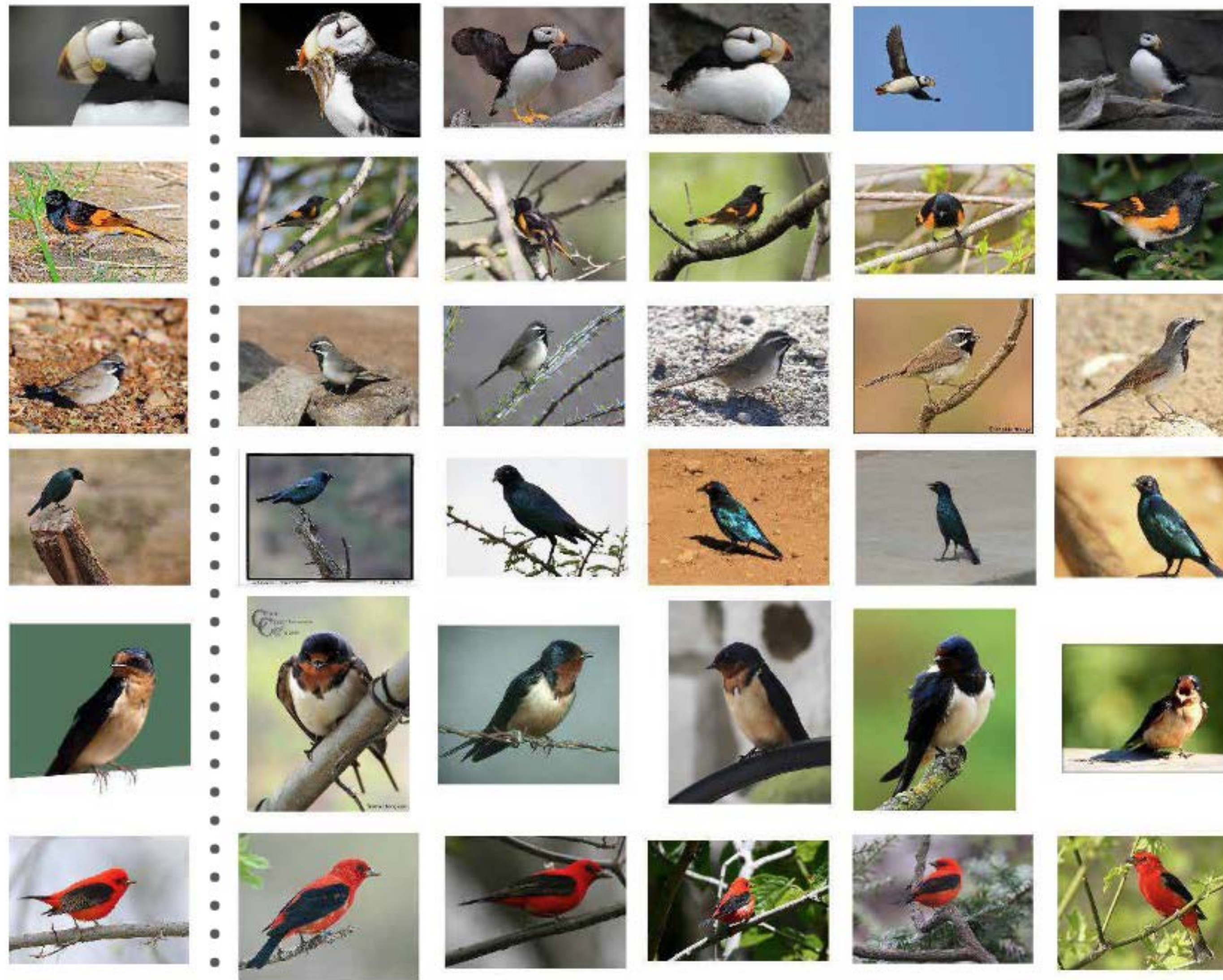


Figure 9: Barnes-Hut t-SNE visualization [36] of our embedding on the test split (class 101 to 200; 5,924 images) of CUB-200-2011. Best viewed on a monitor when zoomed in.

Courtesy of Song, et al. Used under CC BY-NC-SA.

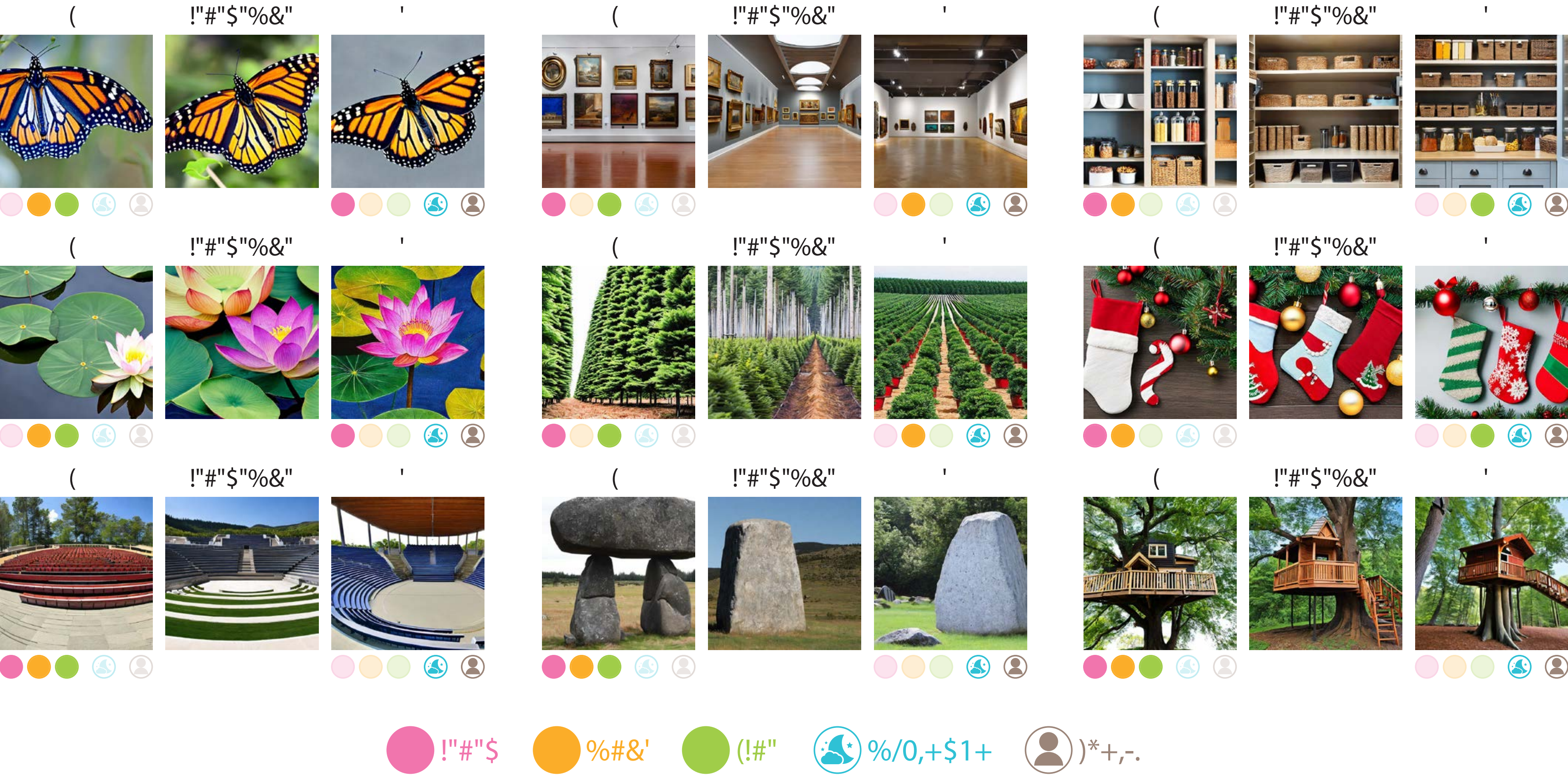
Example query results (neighbors)



Courtesy of Song, et al. Used under CC BY-NC-SA.

figure: Song et al 2015

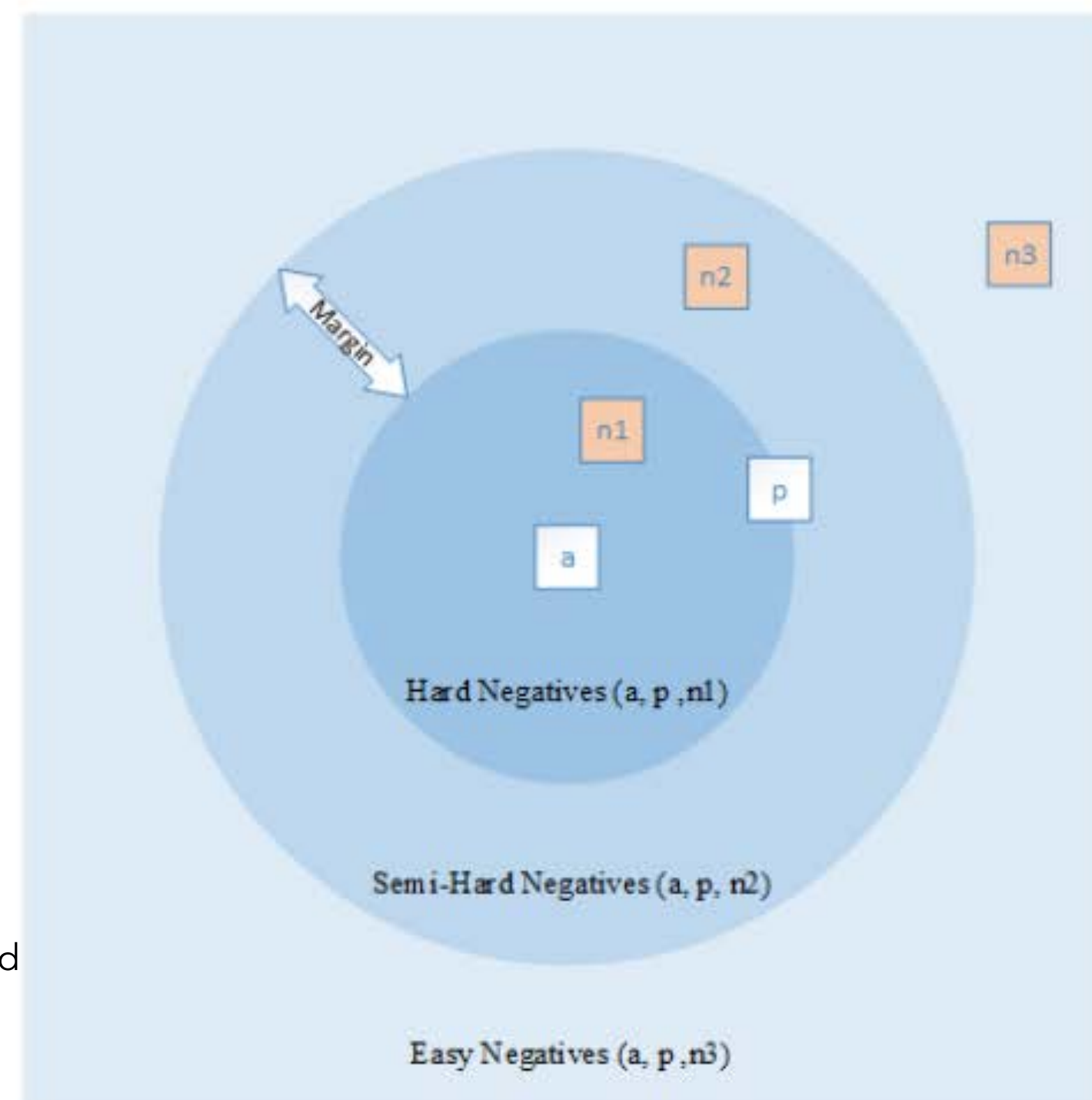
What makes an image “similar”?



Which pairs should we present?

“hard” negatives:

- currently “misplaced”, i.e., closer to anchor than a positive example
- accelerate learning, needed for triplet loss



Hard Negative Mining

$$d(a, n) < d(a, p)$$

Semi-Hard Negative Mining

$$d(a, p) < d(a, n) < d(a, p) + \text{margin}$$

Easy Negative Mining

$$d(a, p) + \text{margin} < d(a, n)$$



Fig 4. Courtesy of Kaya and Bilge. Used under CC BY
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Figure 4. Negative Mining.

Roadmap: similarity-based representation learning

- Representation learning — why?
- What is a “good” representation?
- Metric learning
- Contrastive representation learning (self-supervised)
 - What does it do?
 - Models

Self-supervised contrastive representation learning

- Ideas from metric learning and self-supervision

Common setup

- Encoder maps data onto a hypersphere: $f: \mathcal{X} \rightarrow \mathbb{S}^{d-1}$
- Cross-entropy for softmax "classifier" to discriminate "classes" defined by similarities

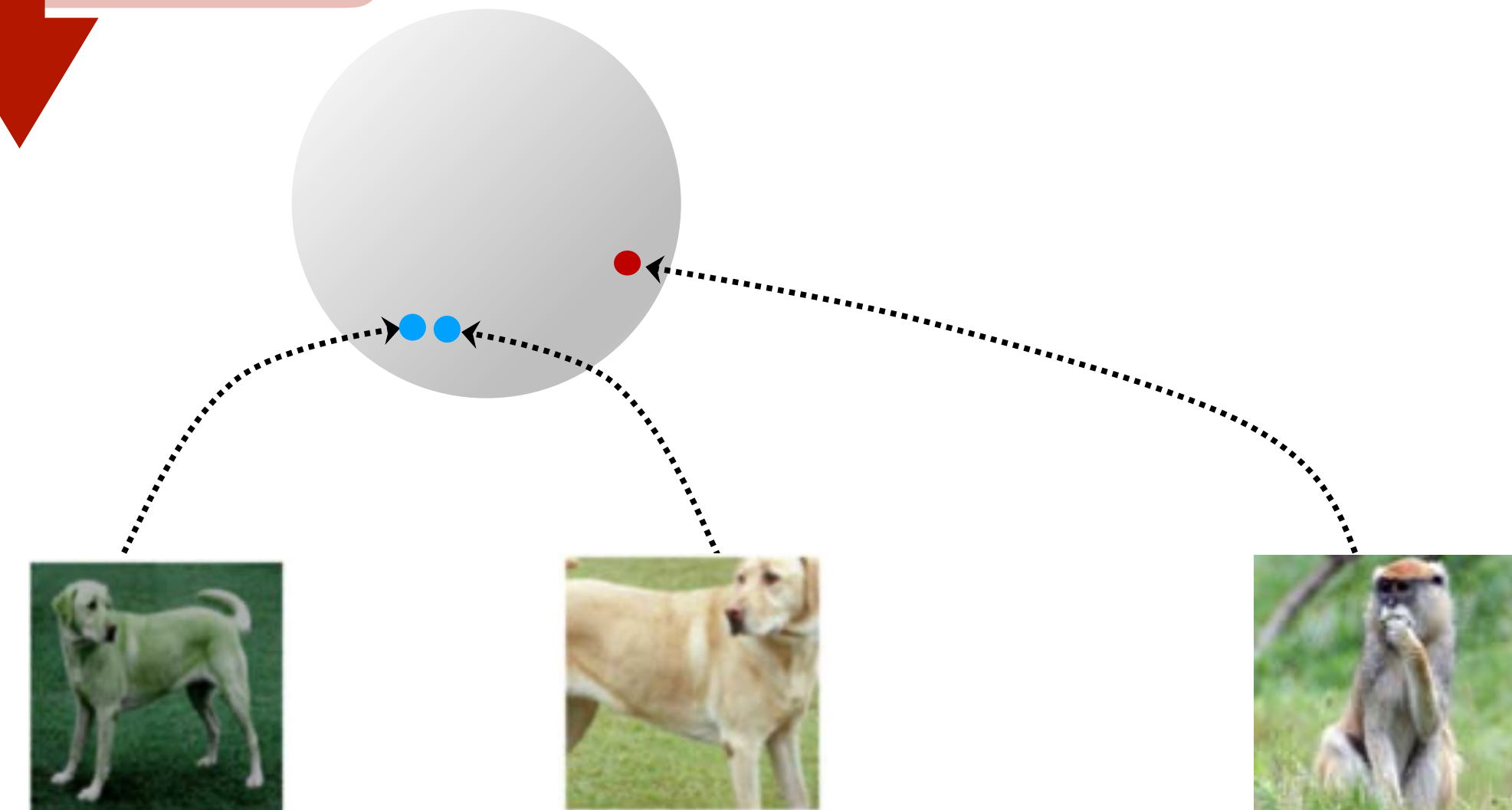
$$\min_f \mathbb{E}_{(\mathbf{x}, \mathbf{x}^+) \sim p_{\text{pos}}, \{\mathbf{x}_i^-\}_{i=1}^N \sim p_{\text{data}}} \left[-\log \frac{e^{f(\mathbf{x}) \cdot f(\mathbf{x}^+) / \sim}}{e^{f(\mathbf{x}) \cdot f(\mathbf{x}^+) / \sim} + \sum_{i=1}^N e^{f(\mathbf{x}) \cdot f(\mathbf{x}_i^-) / \sim}} \right]$$

pull positive pair together

push negative pairs apart

Symmetry: $\forall \mathbf{x}, \mathbf{x}^+, p_{\text{pos}}(\mathbf{x}, \mathbf{x}^+) = p_{\text{pos}}(\mathbf{x}^+, \mathbf{x})$

Matching marginal: $\forall \mathbf{x}, \int p_{\text{pos}}(\mathbf{x}, \mathbf{x}^+) d\mathbf{x}^+ = p_{\text{data}}(\mathbf{x})$



Common setup

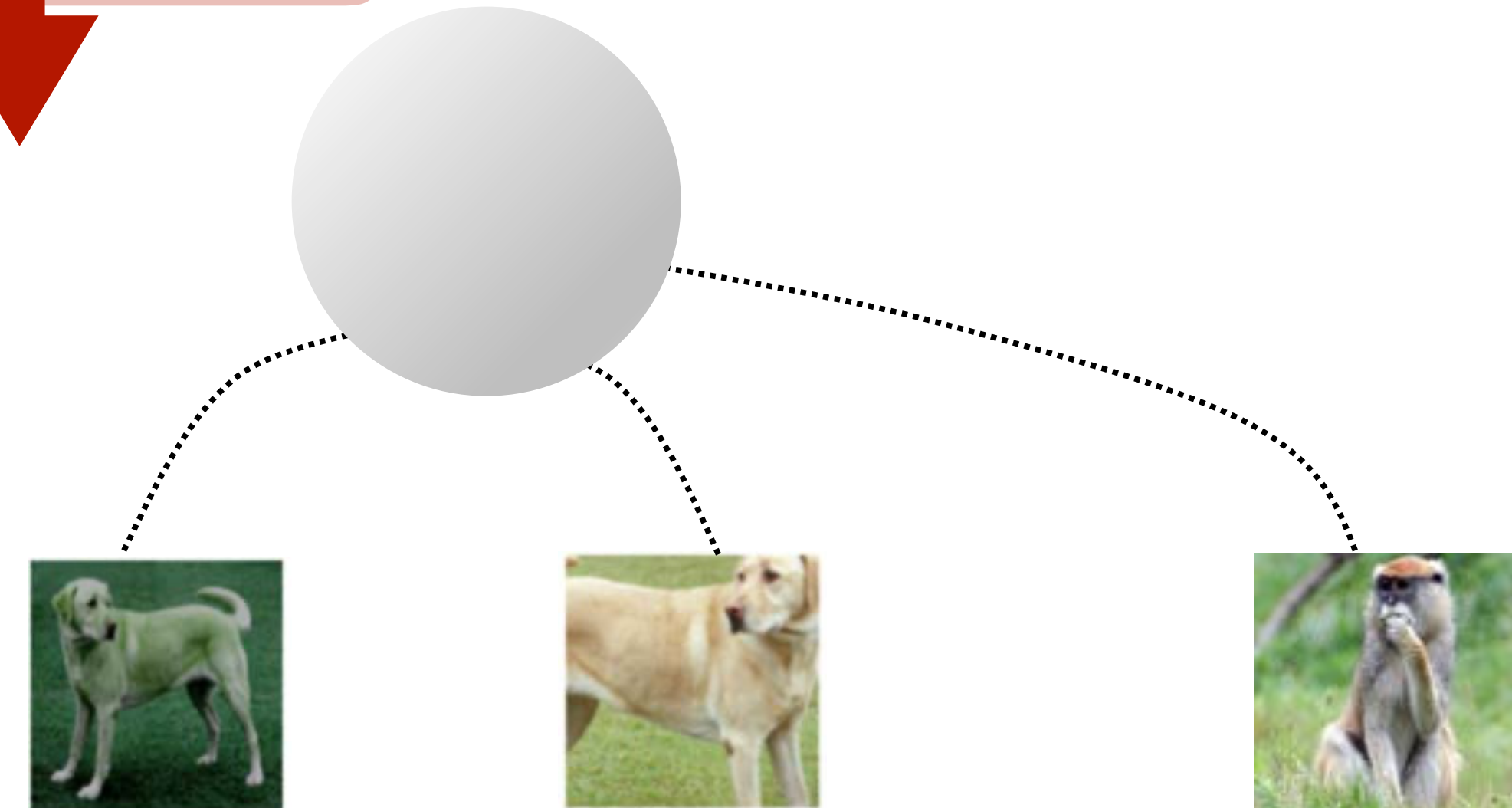
- Encoder maps data onto a hypersphere: $f: \mathcal{X} \rightarrow \mathbb{S}^{d-1}$
- Cross-entropy for softmax "classifier"

$$\min_f \mathbb{E}_{(\mathbf{x}, \mathbf{x}^+) \sim p_{pos}, \{\mathbf{x}_i^-\}_{i=1}^N \sim p_{data}} \left[-\log \frac{e^{f(\mathbf{x}) \cdot f(\mathbf{x}^+) / \tau}}{e^{f(\mathbf{x}) \cdot f(\mathbf{x}^+) / \tau} + \sum_{i=1}^N e^{f(\mathbf{x}) \cdot f(\mathbf{x}_i^-) / \tau}} \right]$$

pull positive pair together

push negative pairs apart

- *Noise-contrastive estimation (NCE) (Gutmann & Hyvärinen 2010), InfoNCE loss (van den Oord et al 2018), ... similar losses also in metric learning*



Common setup

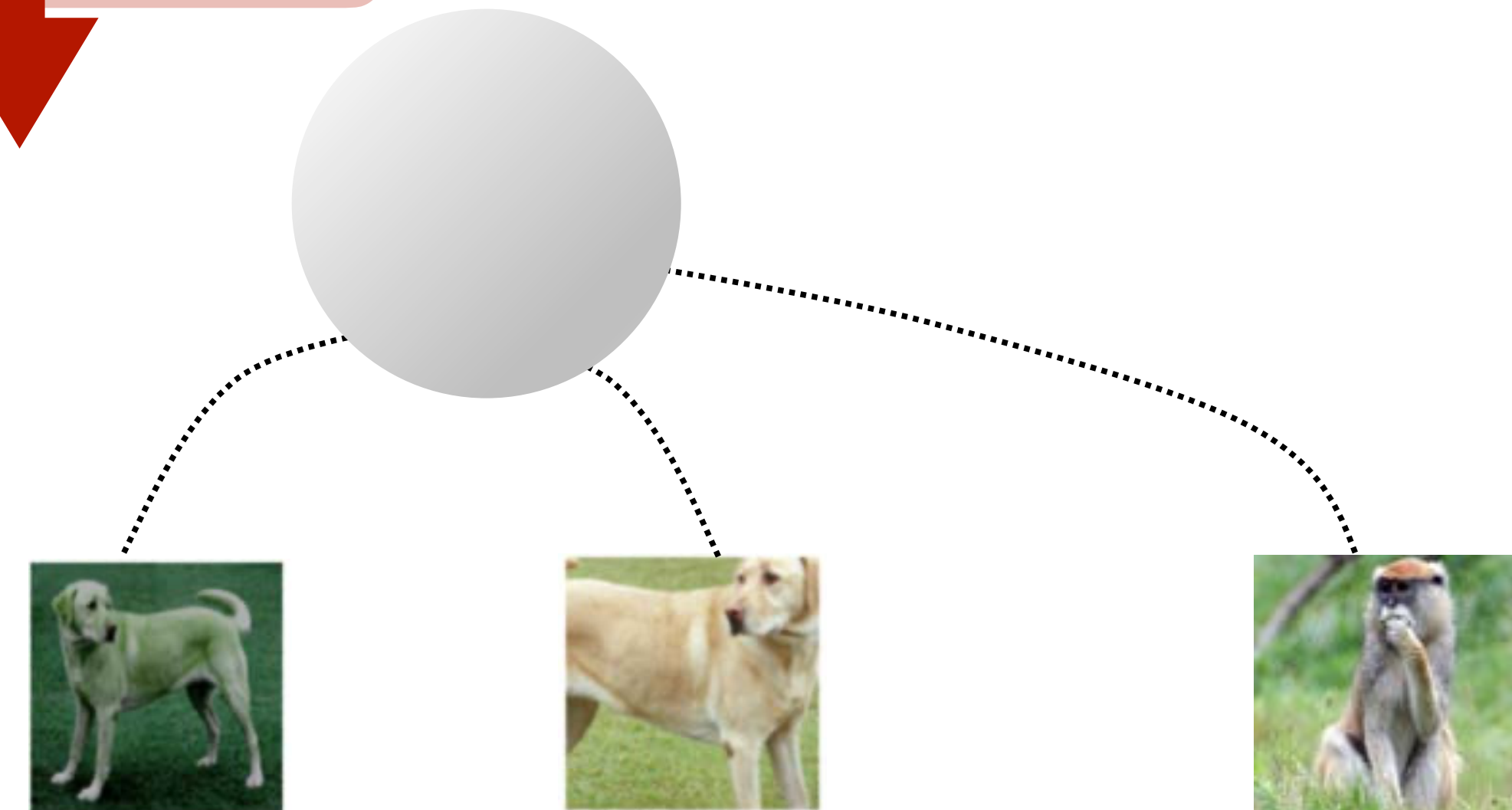
- Encoder maps data onto a hypersphere: $f: \mathcal{X} \rightarrow \mathbb{S}^{d-1}$
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pull positive pair together

push negative pairs apart

As self-supervised learning, can outperform supervised pre-training (for some tasks) (He et al 2020, Misra & van der Maaten 2020)



Why map to a hypersphere?

- more stable training (logistic regression needs regularization)
- well-clustered classes on hypersphere are linearly separable (cut off caps)

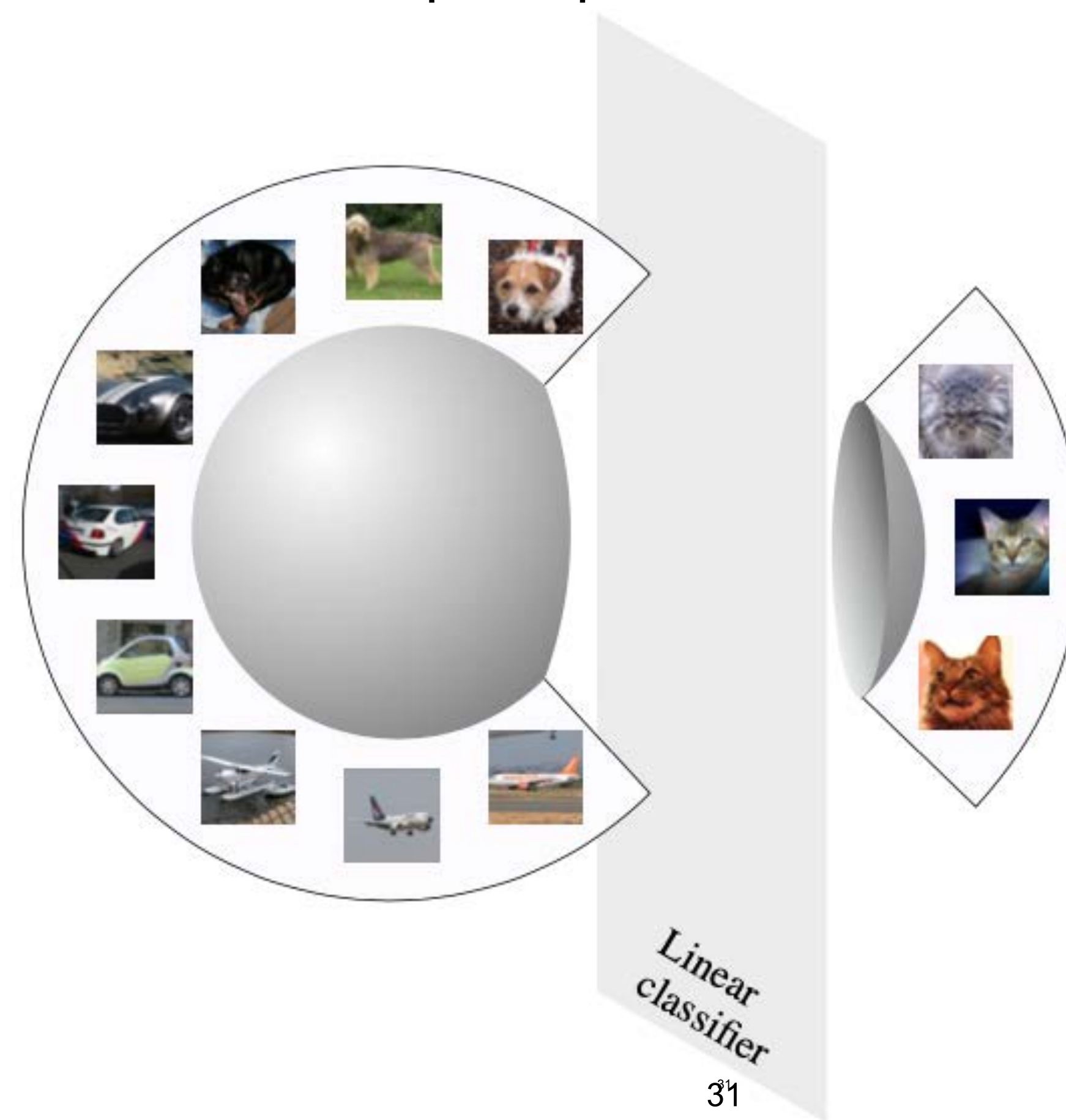


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figure: Wang & Isola 2020

How can we make this “self-supervised”?

$$\min_f \mathbb{E}_{(\mathbf{x}, \mathbf{x}^+) \sim p_{pos}, \{\mathbf{x}_i^-\}_{i=1}^N \sim p_{data}} \left[-\log \frac{e^{f(\mathbf{x}) \cdot f(\mathbf{x}^+) / \tau}}{e^{f(\mathbf{x}) \cdot f(\mathbf{x}^+) / \tau} + \sum_{i=1}^N e^{f(\mathbf{x}) \cdot f(\mathbf{x}_i^-) / \tau}} \right]$$

pull positive pair together

push negative pairs apart

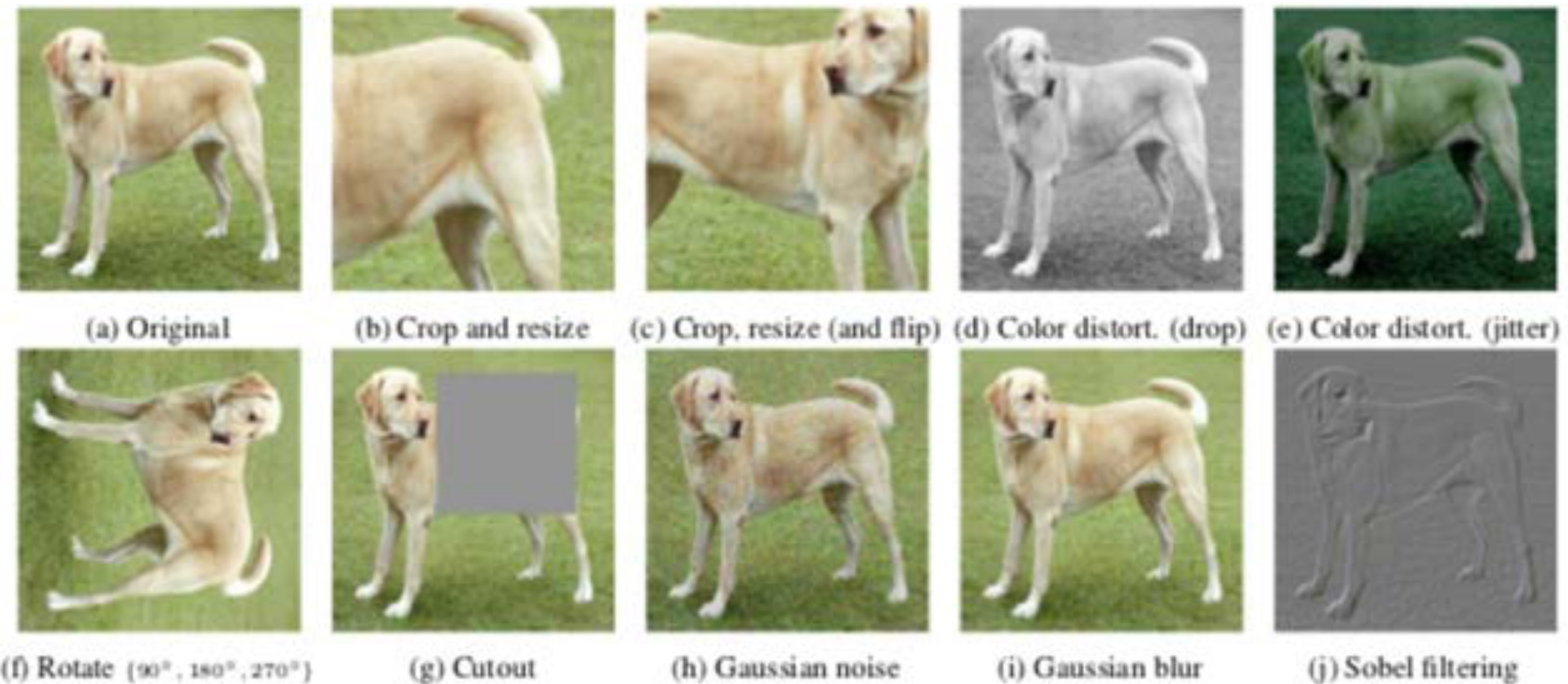
- What are the similar (positive) and dissimilar (negative) pairs?

What are positive and negative examples?

Negative examples:
randomly uniformly
drawn from data



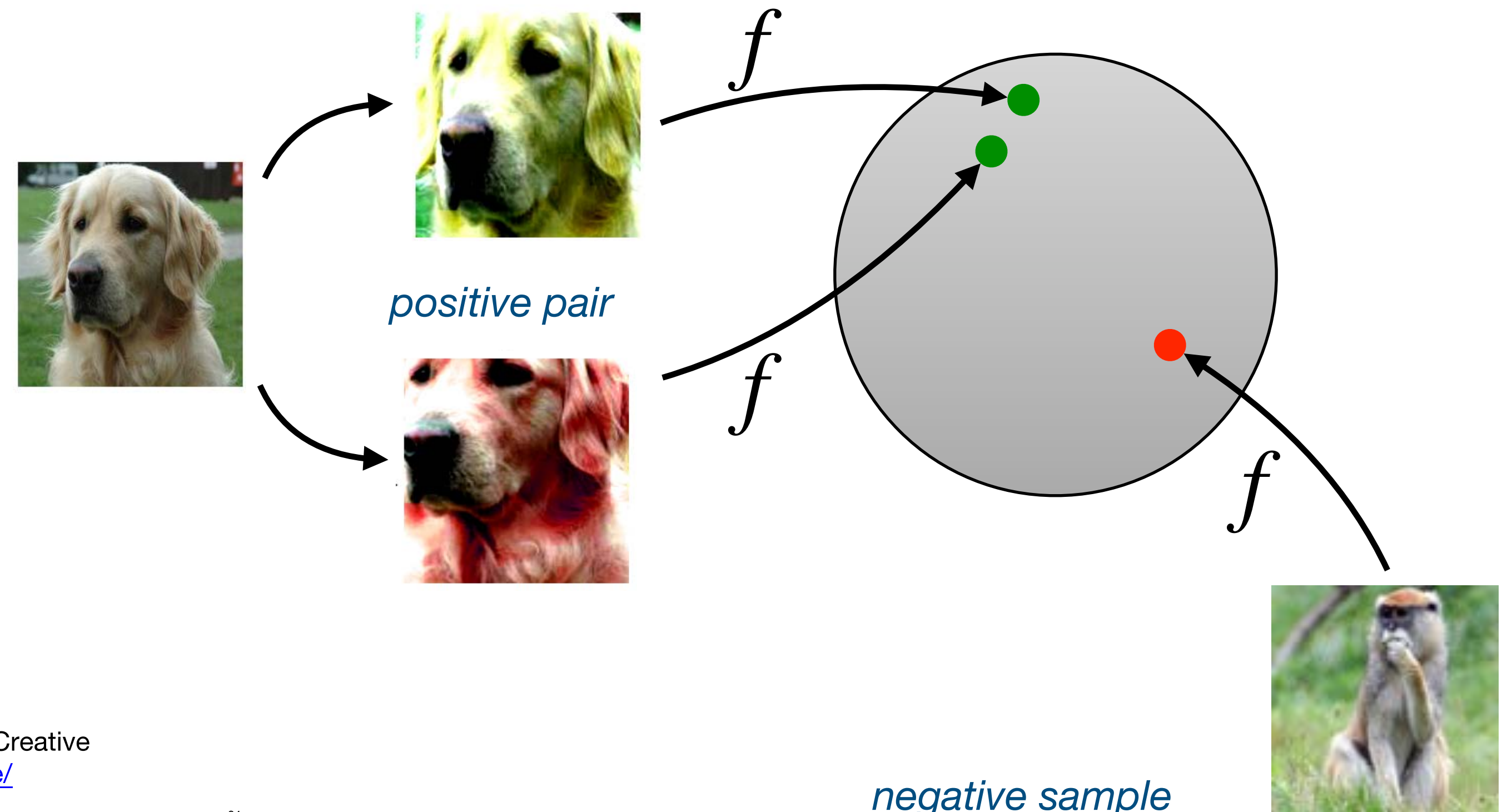
Positive examples:
perturbations that keep
semantic meaning,
data augmentation



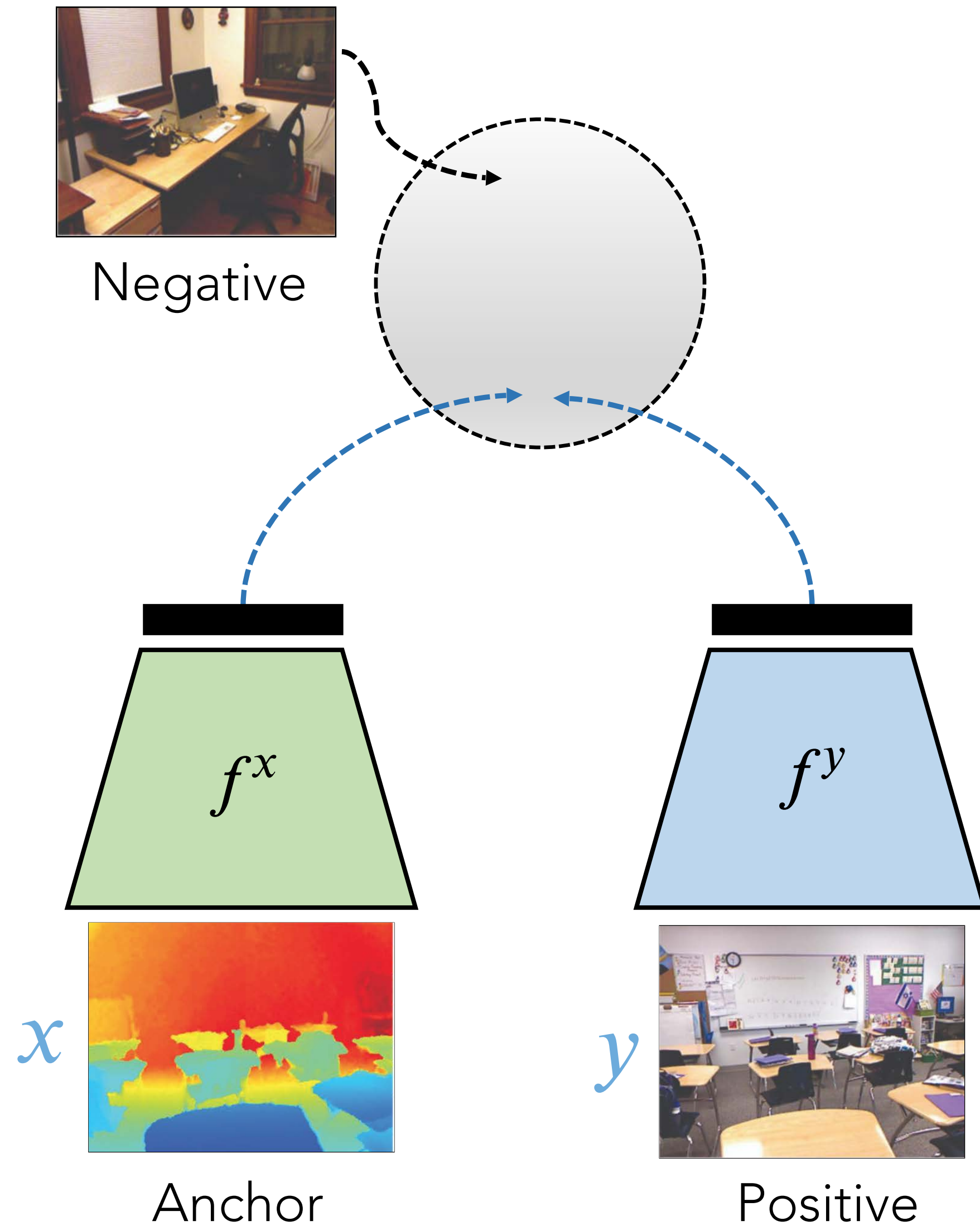
Positive and negative samples

e.g. SimCLR:

- for each data point in the batch, generate 2 random augmentations as positive pair
- all other $2(B-1)$ augmented samples in the batch (of size B) are used as negatives



Variations



(x, y) are two “views” of the same scene

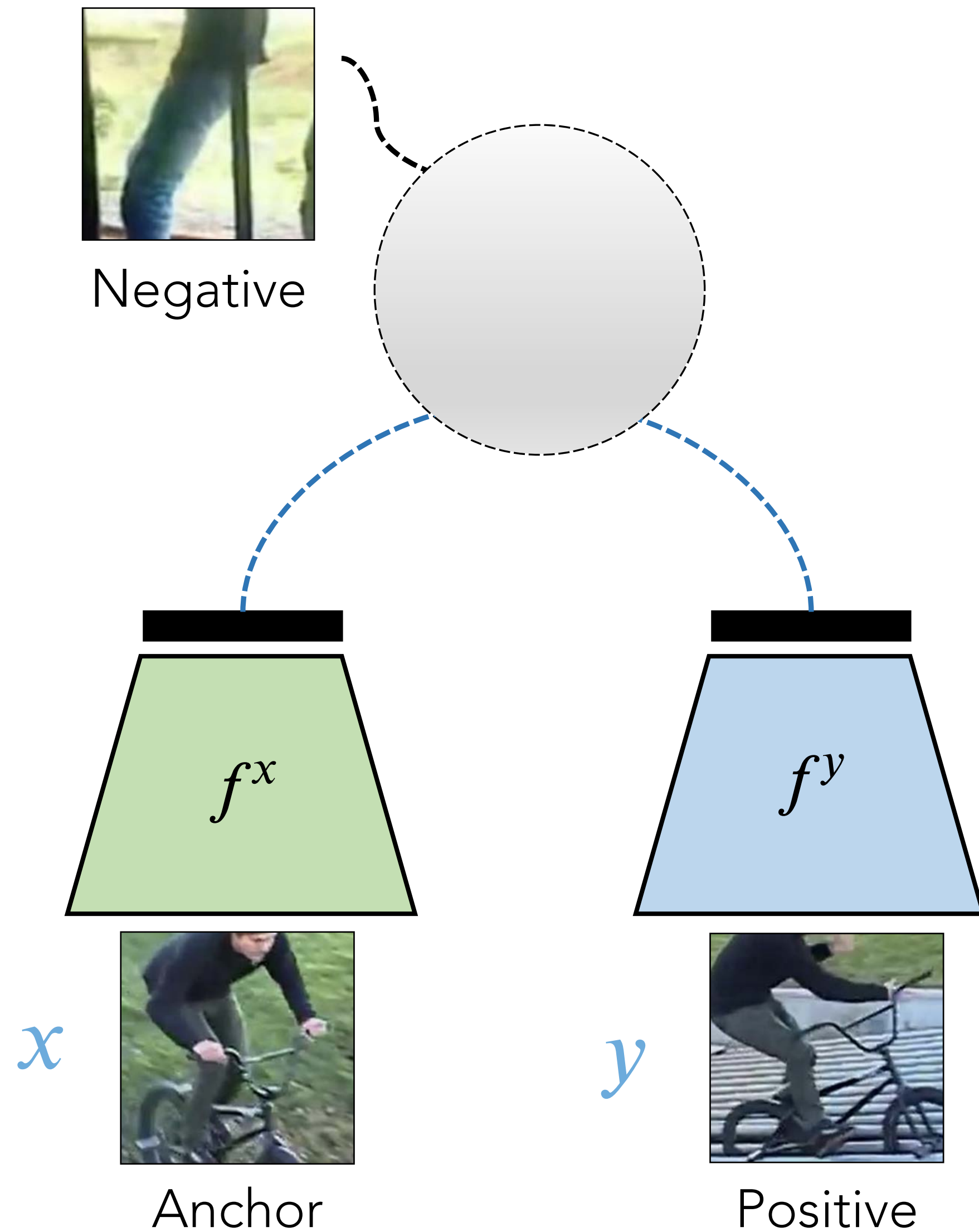
Cross-Channel Representation Learning

[CMC, Tian, Krishnan, Isola 2020]

⋮

Courtesy of Tian, et al. Used under CC BY-NC-SA.

Variations



(x, y) are two “views” of the same scene

Video Representation Learning

[“Slow Feature Learning”, Wiskott & Sejnowski 2002]

[Mobahi, Collobert, Weston 2009]

[Wang & Gupta 2015]

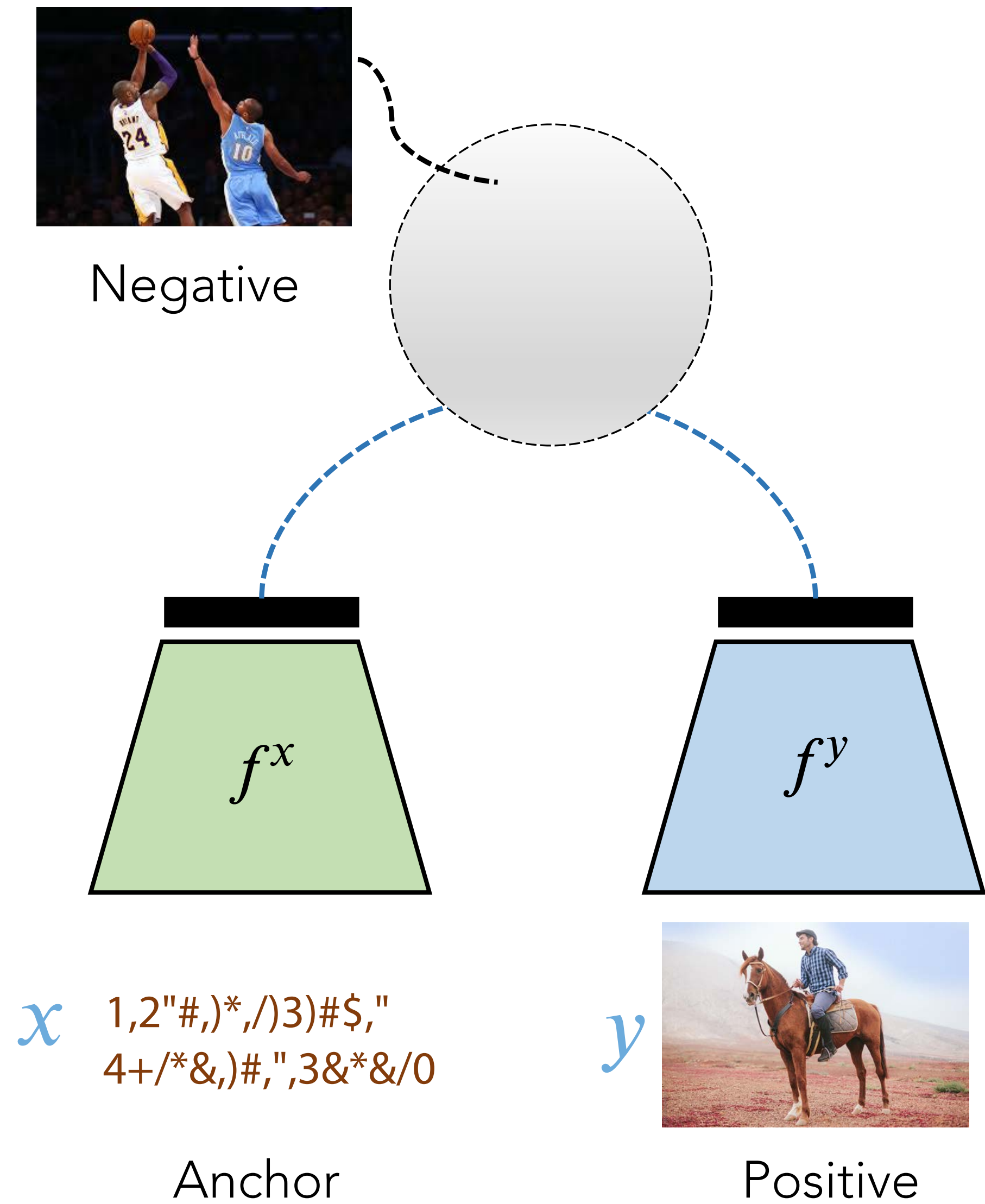
[Isola, Zoran, Krishnan, Adelson 2016]

[Sermanet, Lynch, Chebotar et al. 2018]

[van den Oord, Li, Vinyals 2018]

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Variations



$(x, y)!$

!"#\$%&'()*+,-./:;<=>?@A

[Karpathy, Joulin, Fei-Fei 2014]

⋮

[CLIP, Radford, Kim et al. 2021]

What is this method doing?

2 ingredients:

- Contrastive loss (which specific form)
- Data (which positive/negative pairs)

What is the contrastive loss doing?

$$\mathcal{L}_{cont}(f) = \mathbb{E}_{(\mathbf{x}, \mathbf{x}^+) \sim p_{pos}, \{\mathbf{x}_i^-\}_{i=1}^N \sim p_{data}} \left[-\log \frac{e^{f(\mathbf{x}) \cdot f(\mathbf{x}^+) / \tau}}{e^{f(\mathbf{x}) \cdot f(\mathbf{x}^+) / \tau} + \sum_{i=1}^N e^{f(\mathbf{x}) \cdot f(\mathbf{x}_i^-) / \tau}} \right]$$

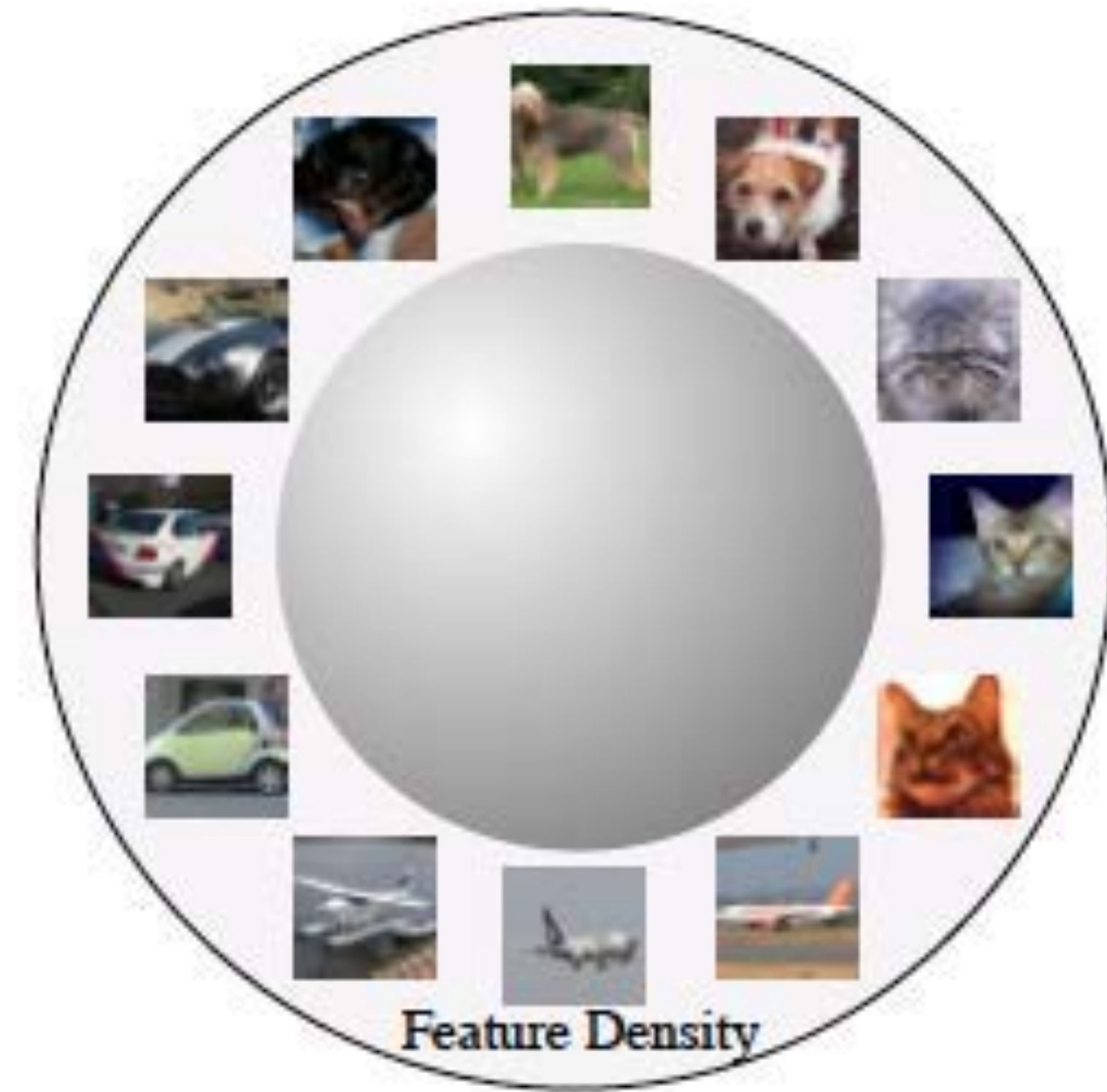
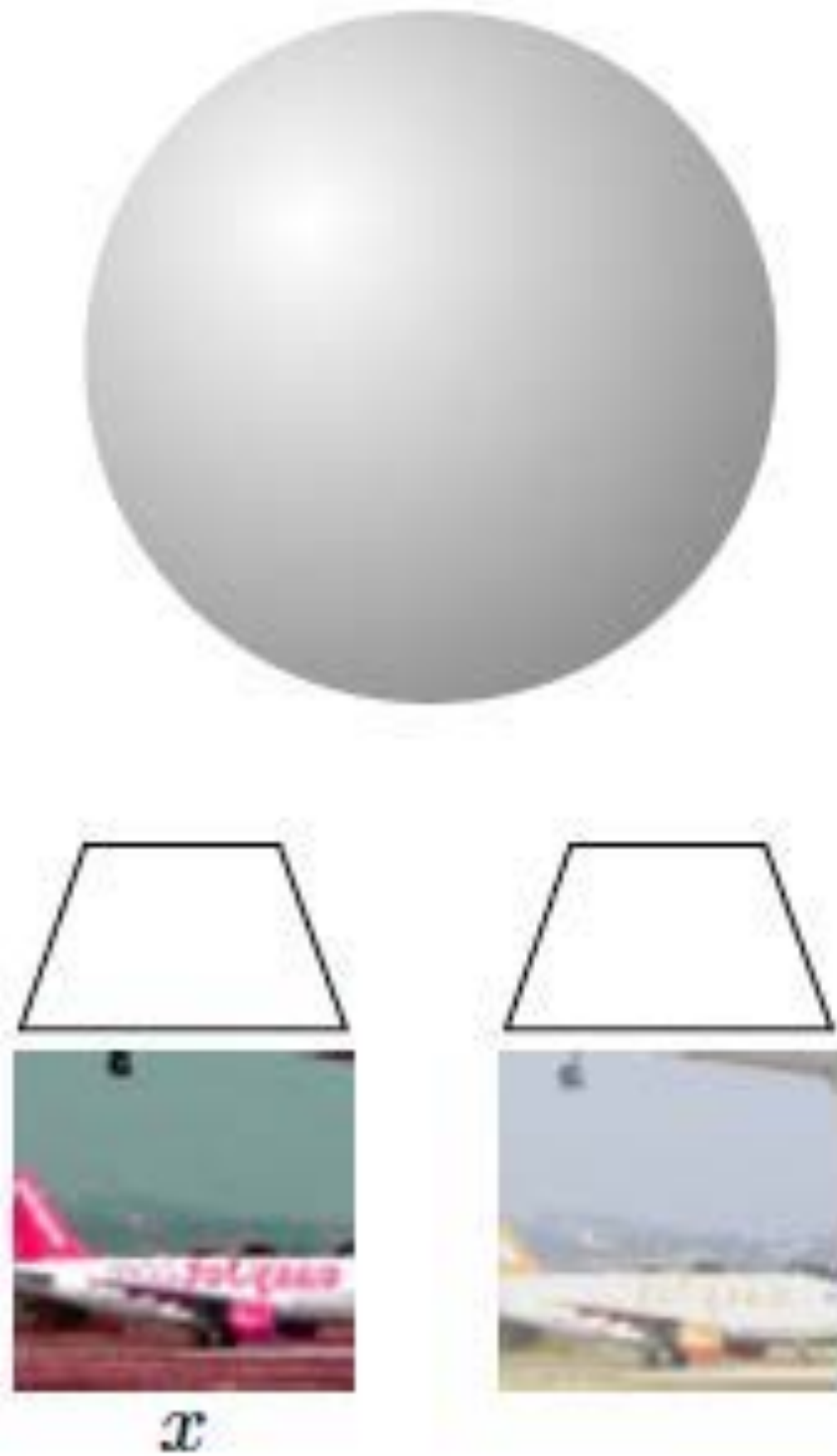
- cross-entropy loss to distinguish data points
- maximizes a lower bound on mutual information between “views” $f(\mathbf{x}), f(\mathbf{x}^+)$ (Poole et al, 2019):

$$\text{MI}(f(\mathbf{x}), f(\mathbf{x}^+)) \geq \log(N) - \mathcal{L}_{cont}(f)$$

What (else) is the contrastive loss doing?

- Recall: properties of “good” representations:
 1. **Concentration/Alignment**: Data from the same class is close together, remove irrelevant information
 2. **Separation**: classes are well separated, do not lose information
 3. **Robustness** to irrelevant perturbations

Alignment and separation

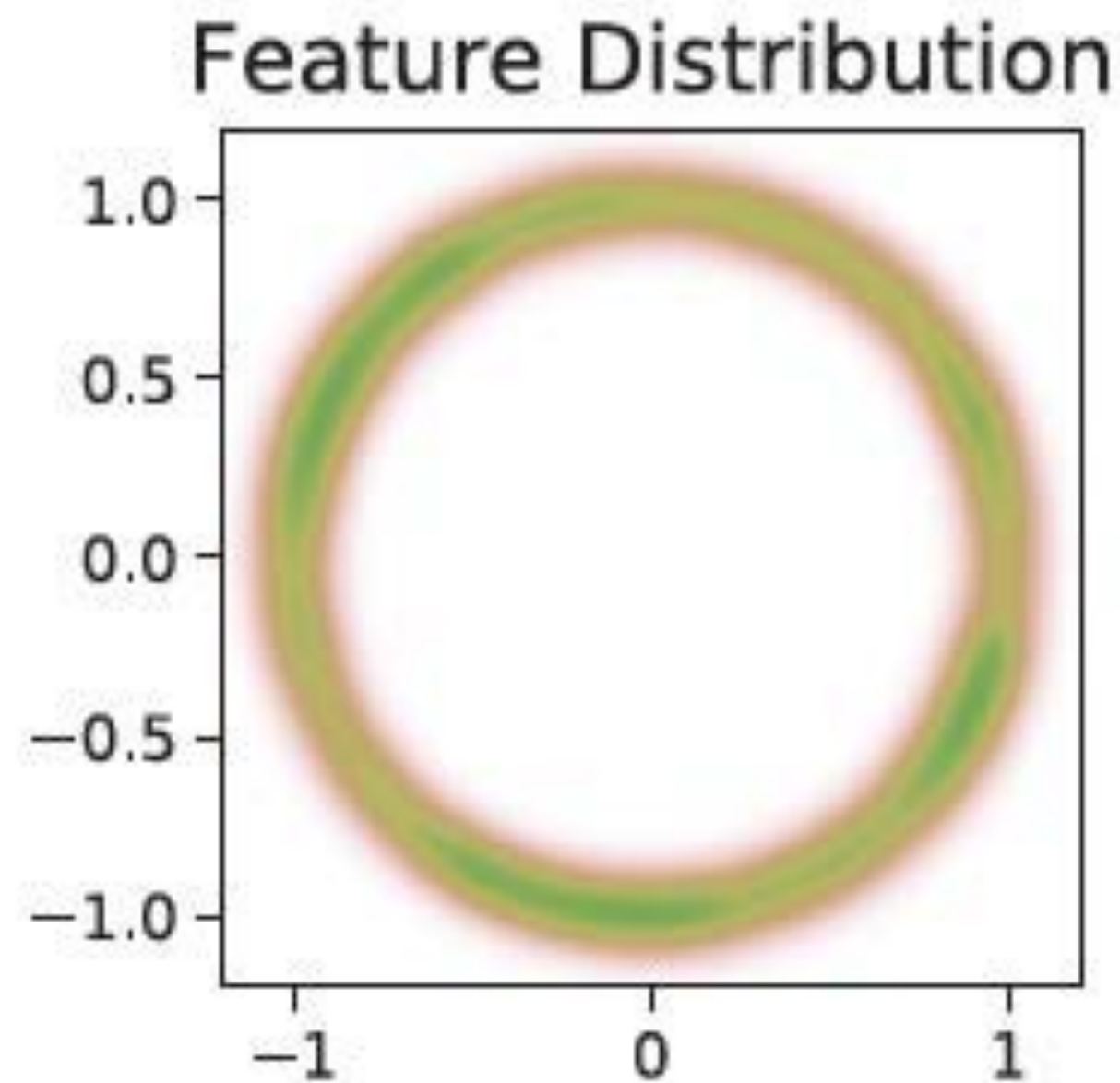


Uniformity: Preserve maximal information

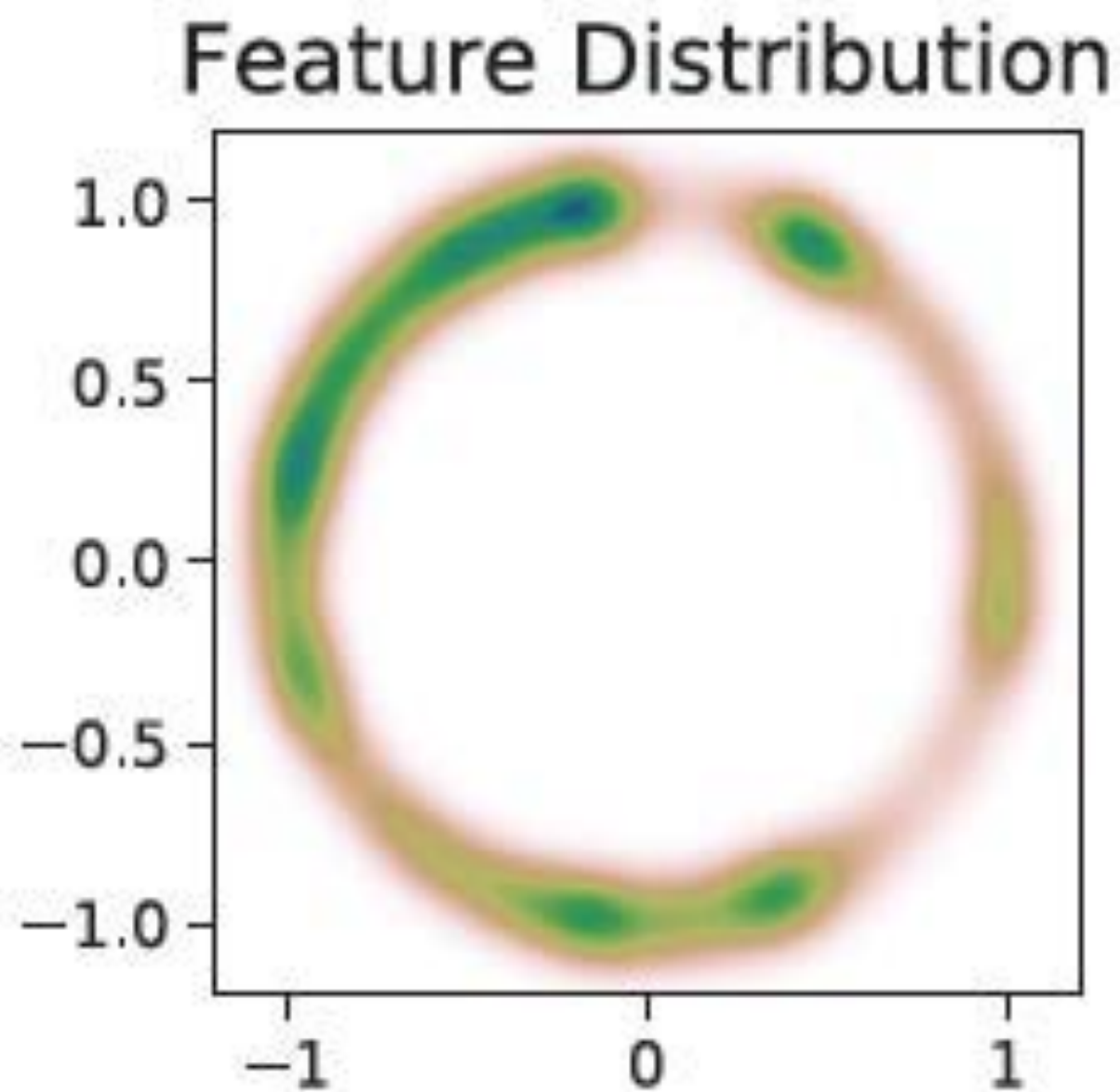
Feature distribution from Contrastive Learning

Toy example:

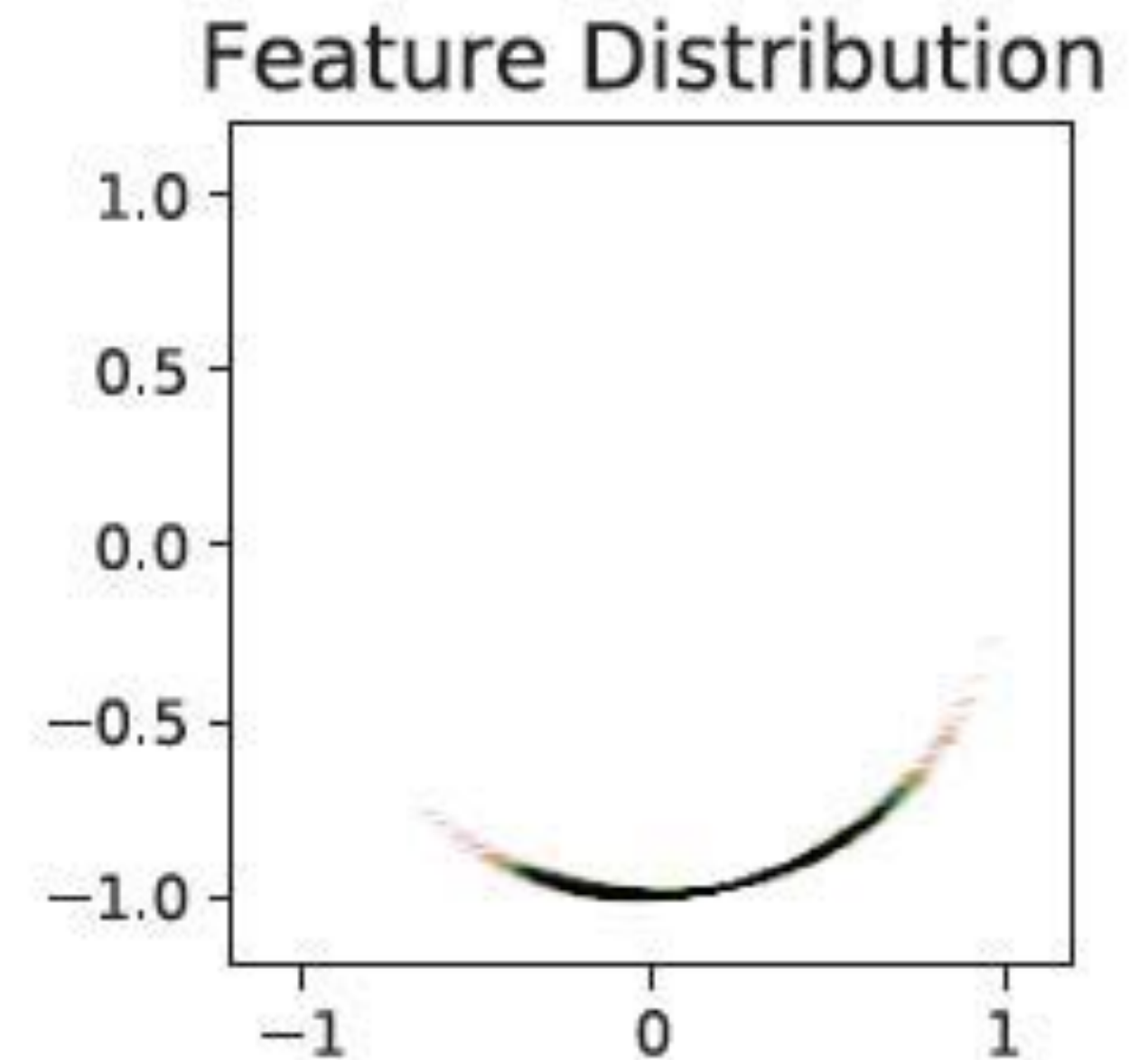
Train CIFAR-10 encoders with $\mathcal{S}^1$ feature space (circle).
Visualize feature distributions on the validation set.



Unsupervised Contrastive
Learning

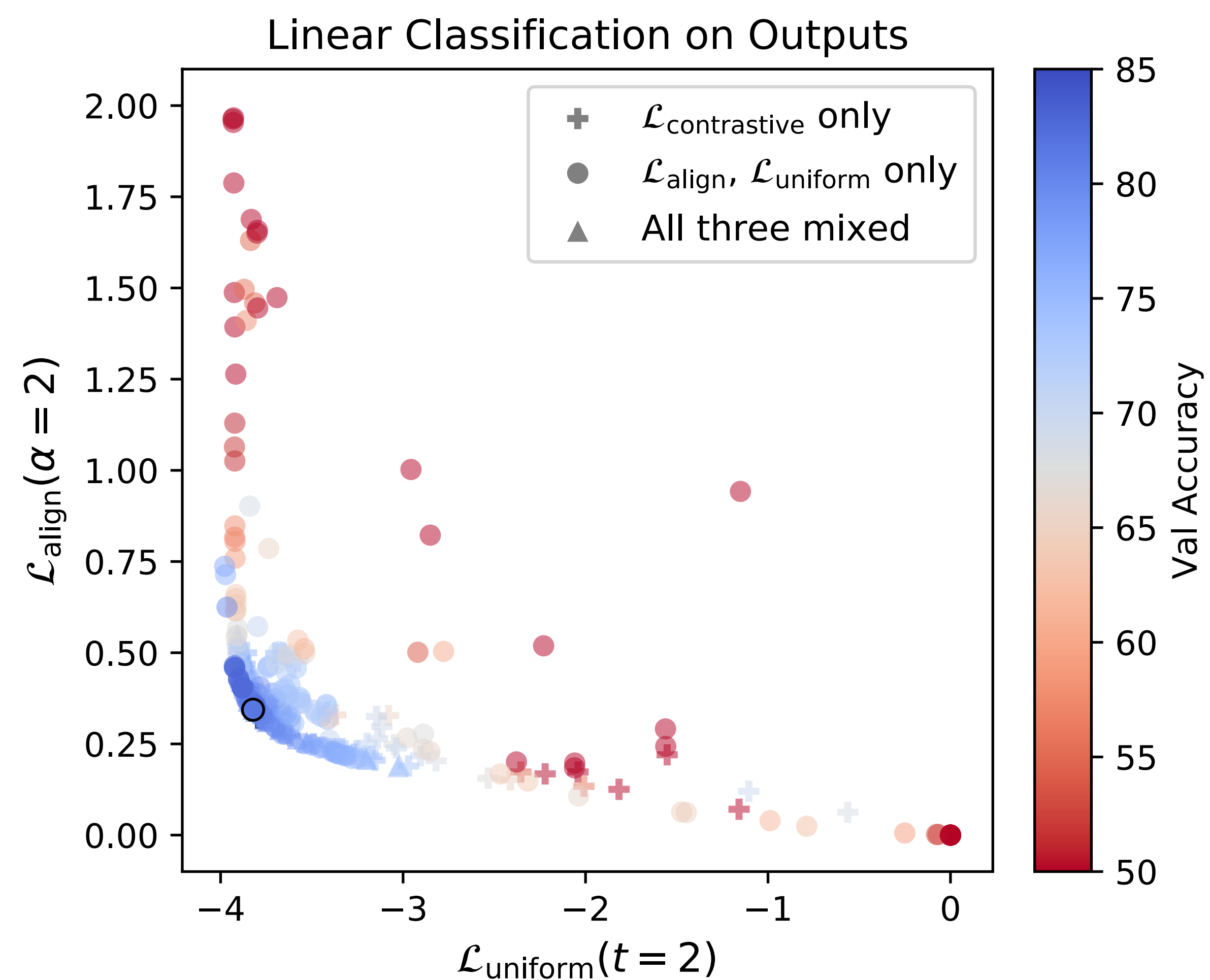


Supervised Predictive
(NLL) Learning

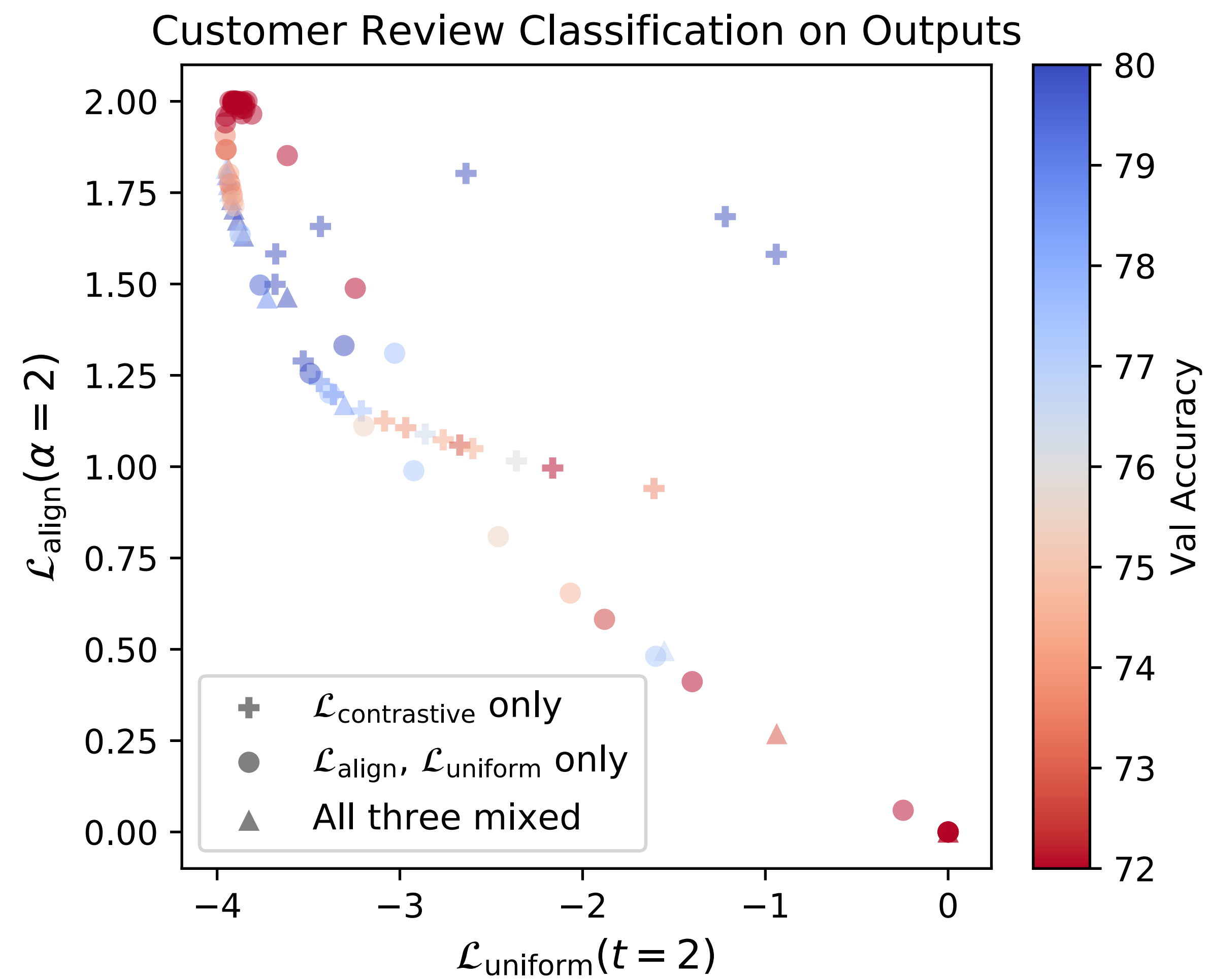


Random Network
Initialization

Relation Between Representation Quality and Alignment & Uniformity



306 STL-10 Encoders



108 BookCorpus Encoders

What is the contrastive loss doing?

- Loss function encourages:
 1. **Concentration/Alignment**: Data from the same class is close together, remove irrelevant information
 2. **Separation**: classes are well separated, do not lose information
- What do the selection of positive and negative pairs encourage?

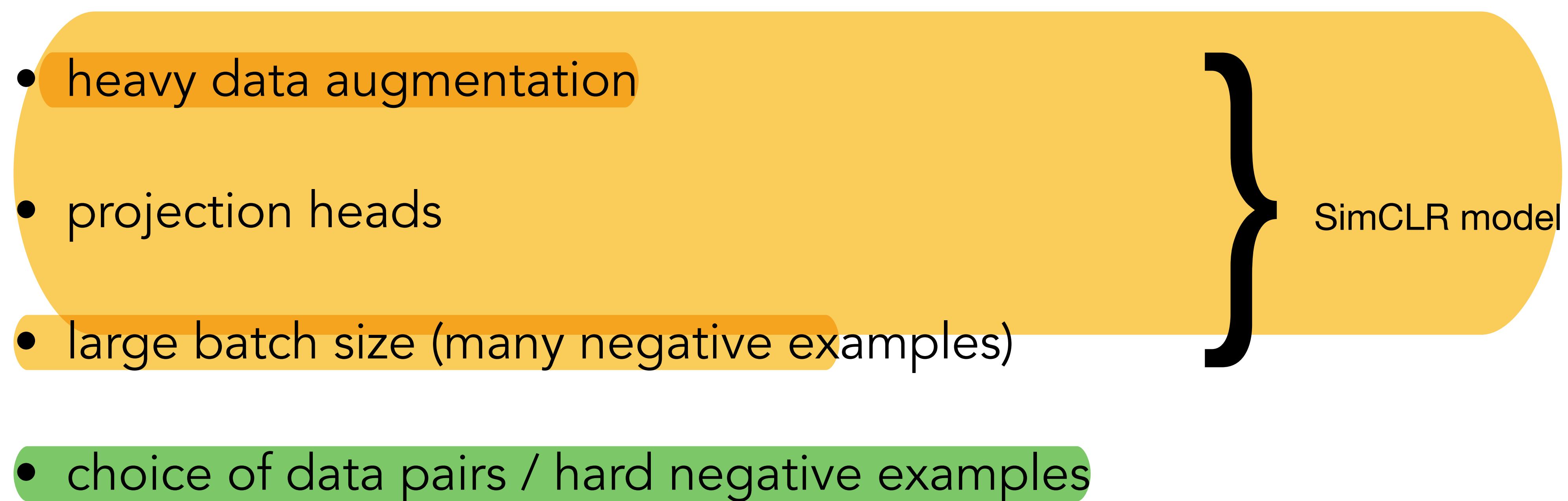
What are we “teaching” the model via choice of pairs?

- positive pairs = augmentations of the same data point should be close
- => learned representation is invariant to perturbations induced by data augmentations: **learned invariance**
- Finding the “right” invariances can be challenging for different types of data
- Learned versus hard-coded invariances (geometric DL lecture): when would we use which?

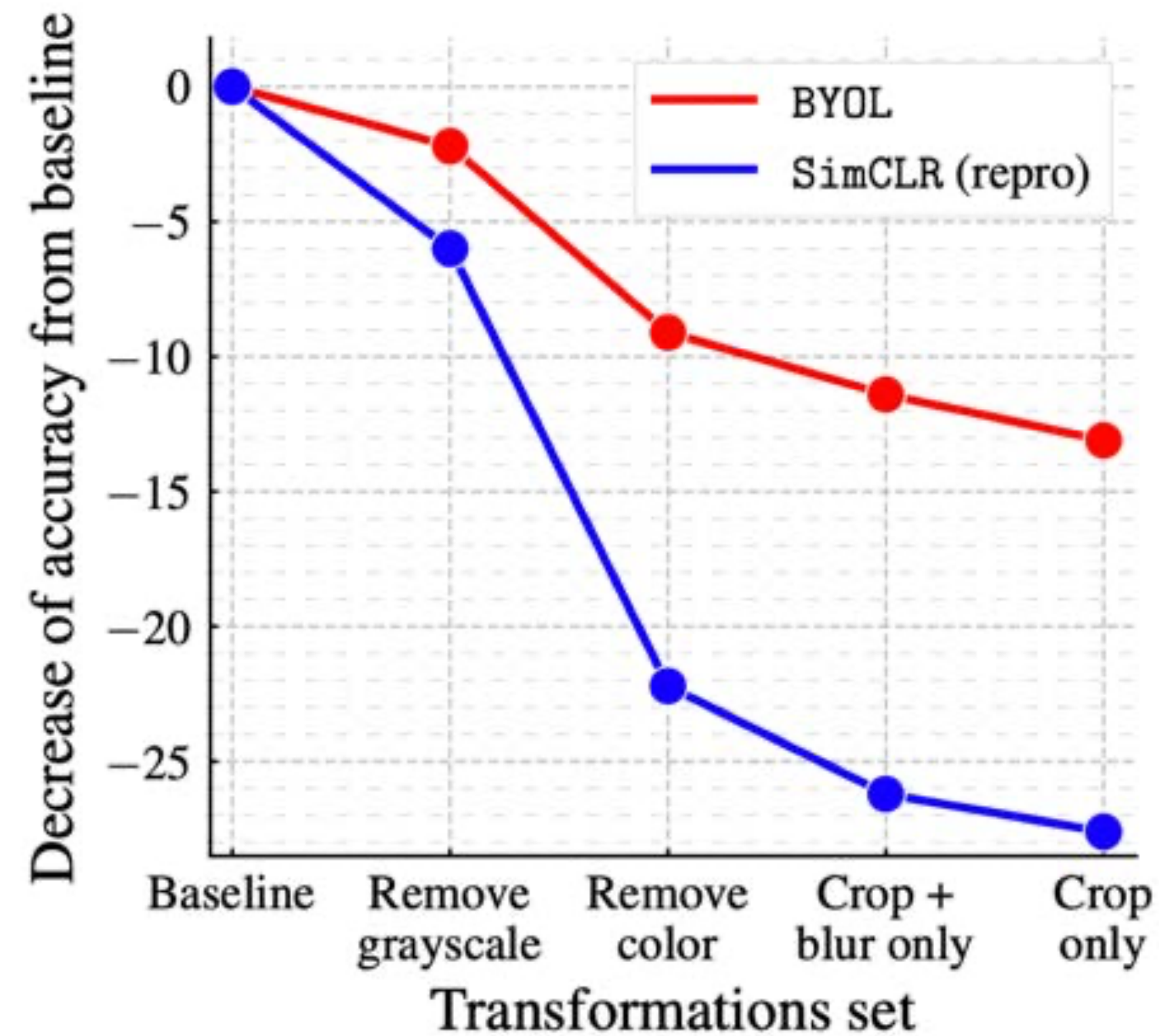
What is the contrastive loss doing?

- Loss function encourages:
 1. **Concentration/Alignment**: Data from the same class is close together, remove irrelevant information
 2. **Separation**: classes are well separated, do not lose information
- Data encourages:
 3. **Robustness** to **irrelevant** perturbations

Ingredients to make self-supervised CL work (better)

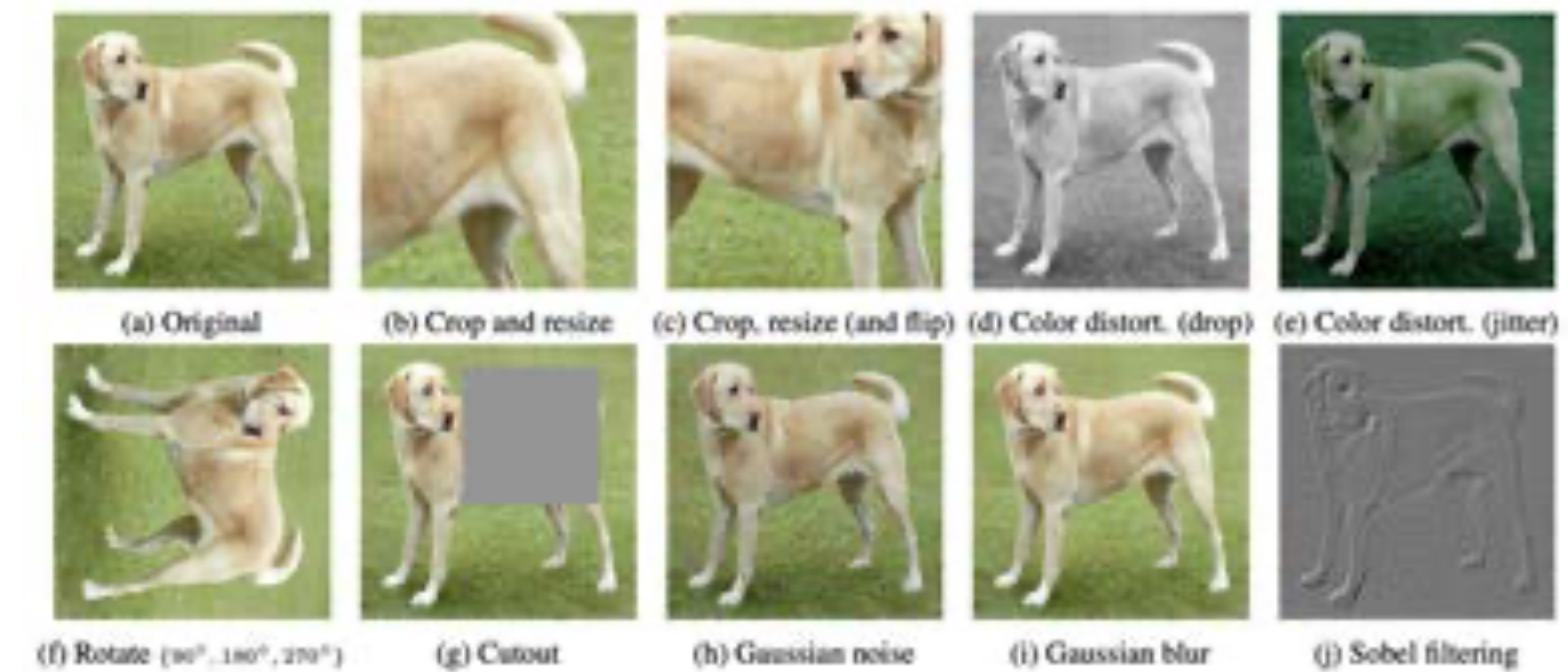
- heavy data augmentation
 - projection heads
 - large batch size (many negative examples)
 - choice of data pairs / hard negative examples
- 
- SimCLR model

Effect of data augmentation



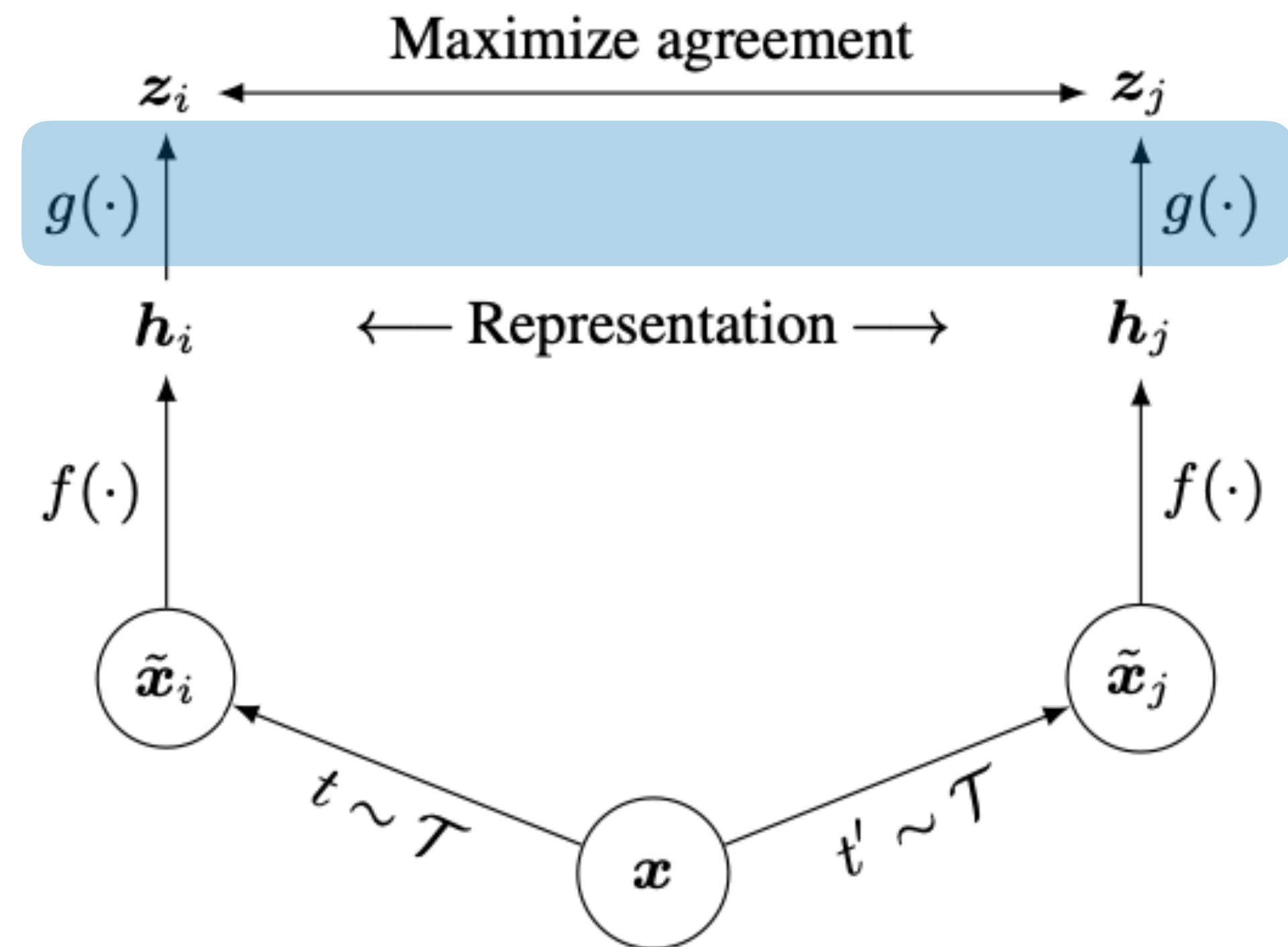
Impact of progressively removing transformations

(figure: Grill et al 2020)



Projection head

- contrastive loss is applied to a transformed version $g(\mathbf{h})$ of the representation $\mathbf{h}$
- g is linear or small MLP
- use $\mathbf{h}$ for downstream task
- Projection head improves performance!

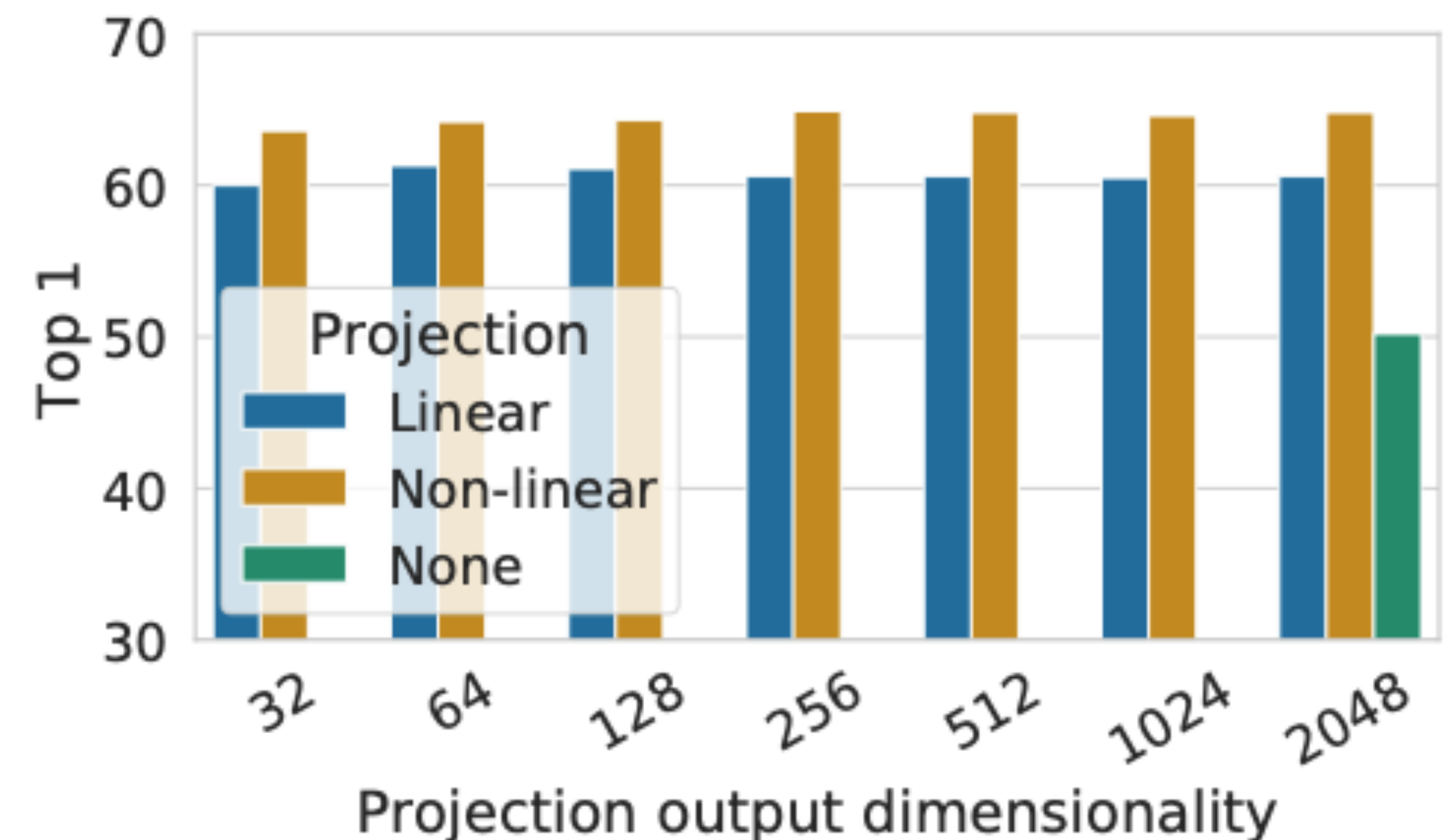
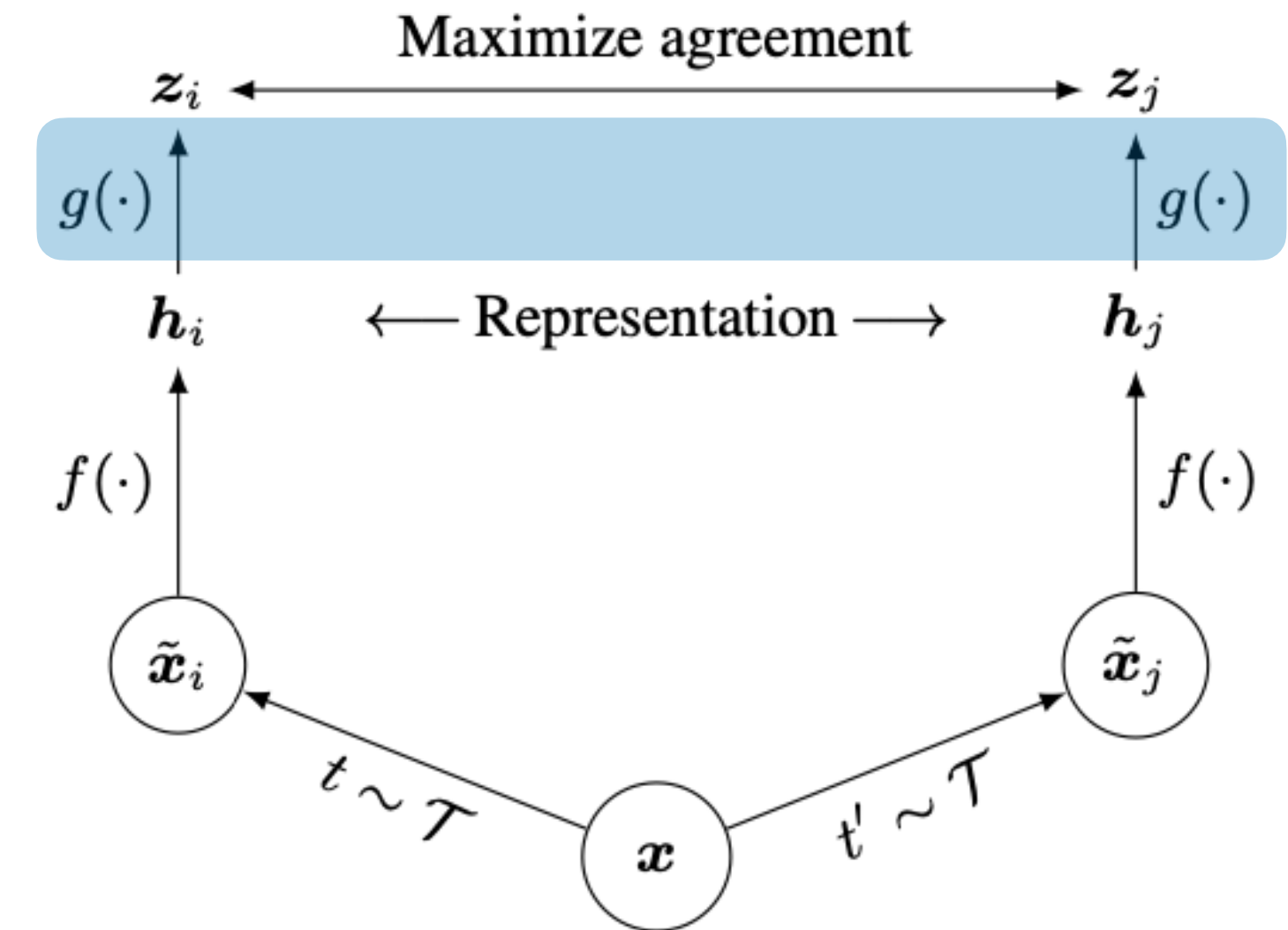


Projection head

- Projection head improves performance.

- Why?

Possibly because representation $\mathbf{h}$ then need not be completely invariant to augmentations, can retain some information



Effect of batch size

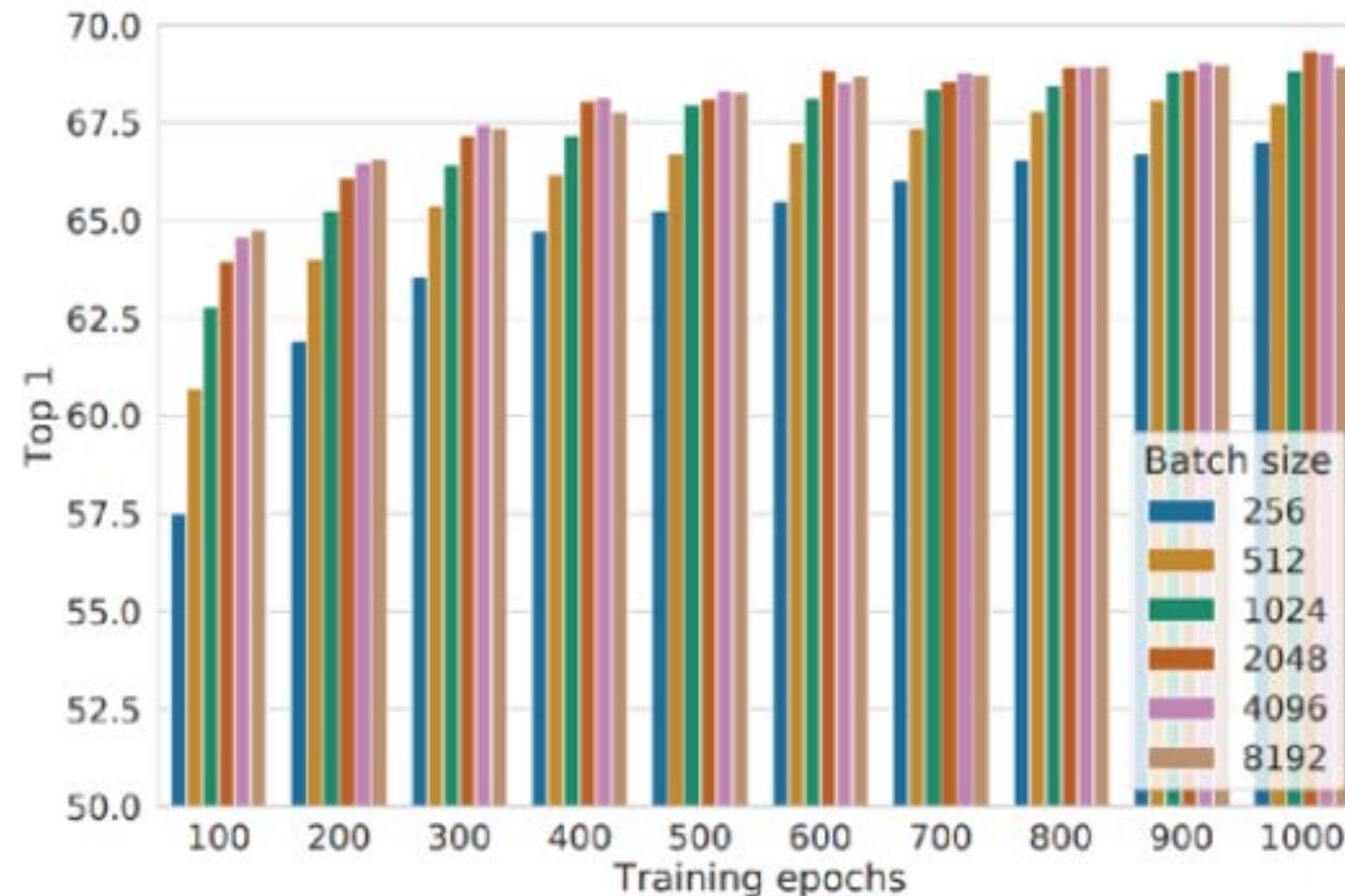


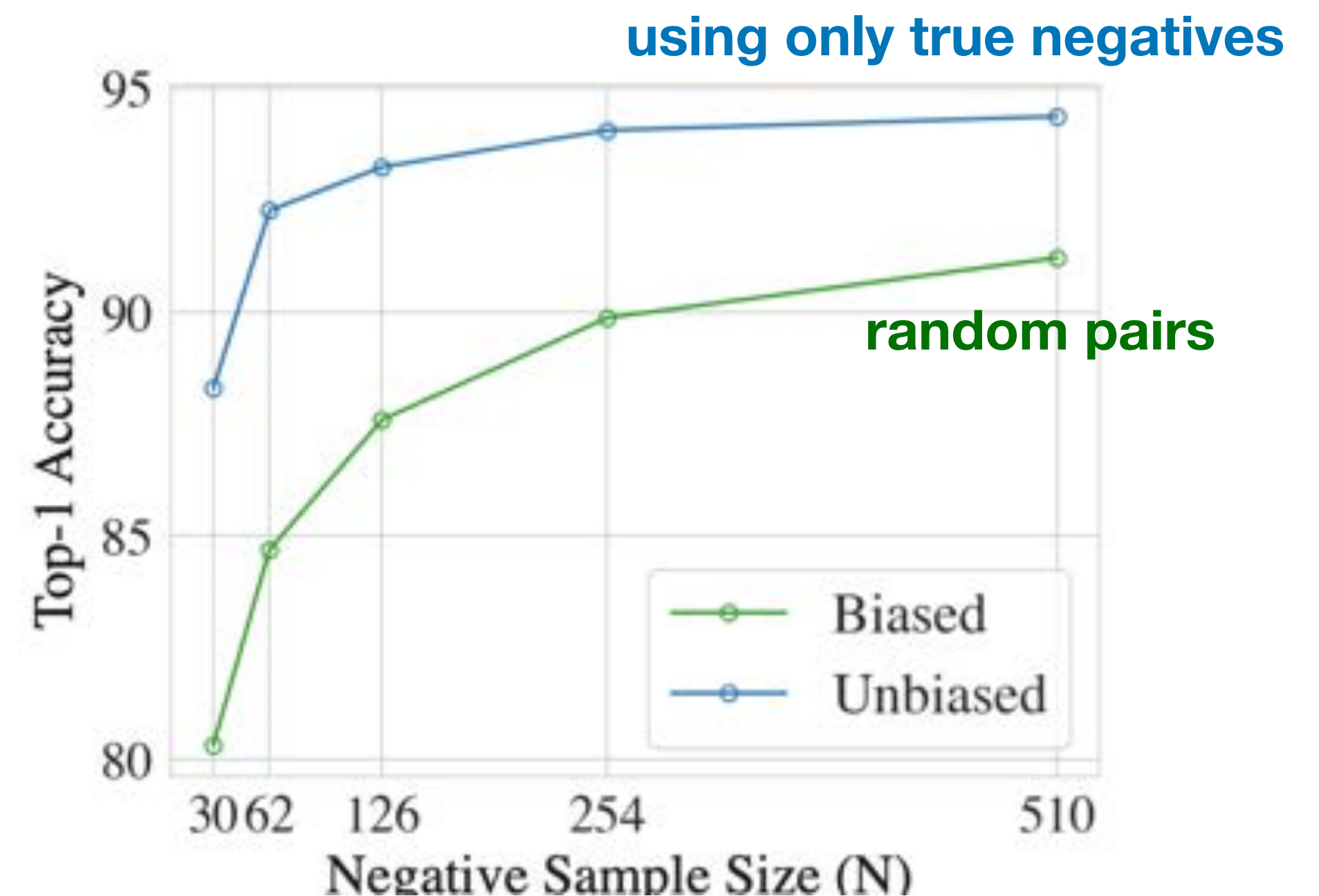
Figure 9. Linear evaluation models (ResNet-50) trained with different batch size and epochs. Each bar is a single run from scratch.¹⁰

(Figure from Chen et al. 2020)

- SimCLR uses all points in a batch as negative examples for a positive pair
- needs large number of negative pairs = large batch sizes
- Expensive. Newer methods make this more efficient (like MoCo, He et al. 2020)

Improving negative samples

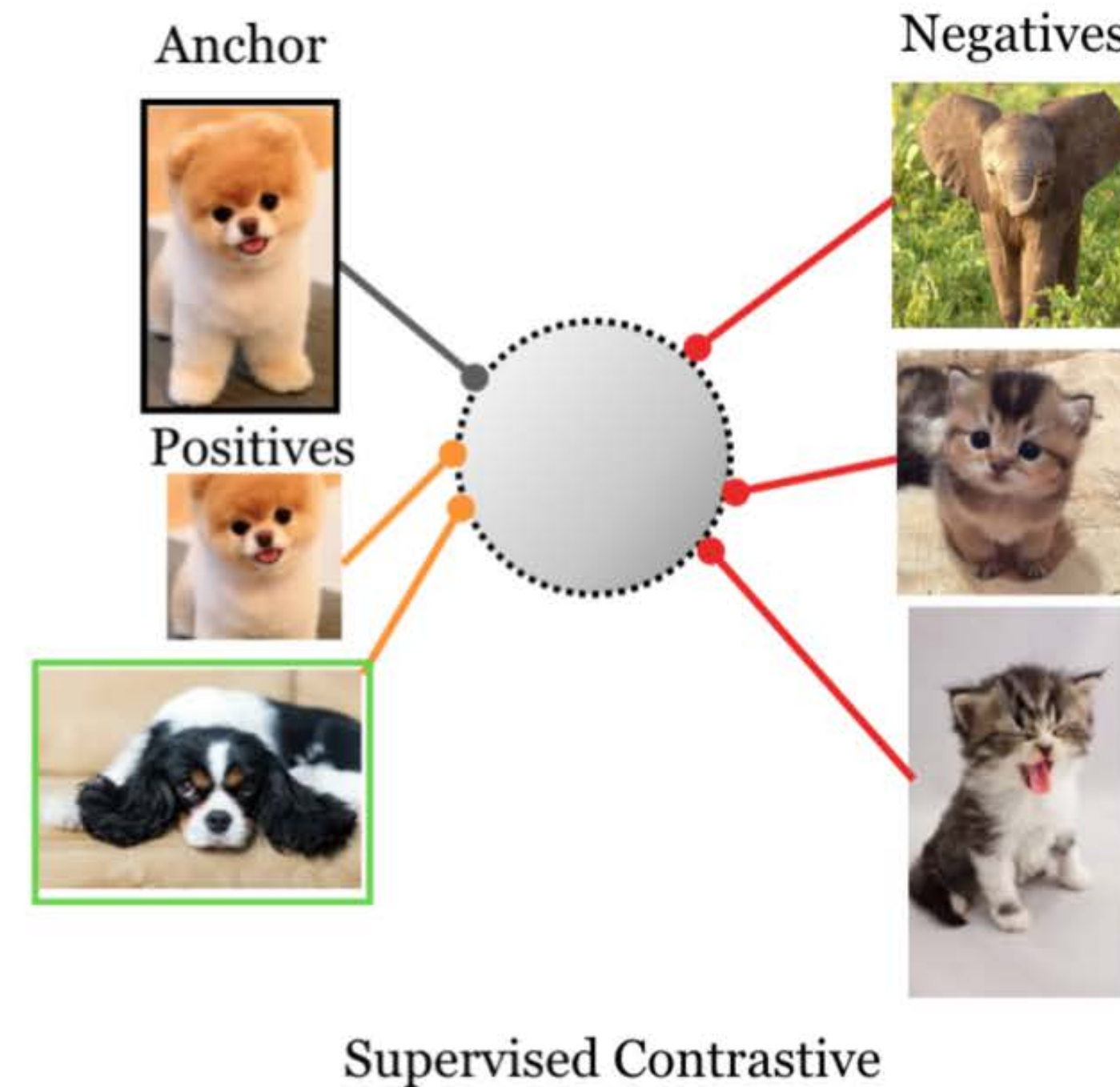
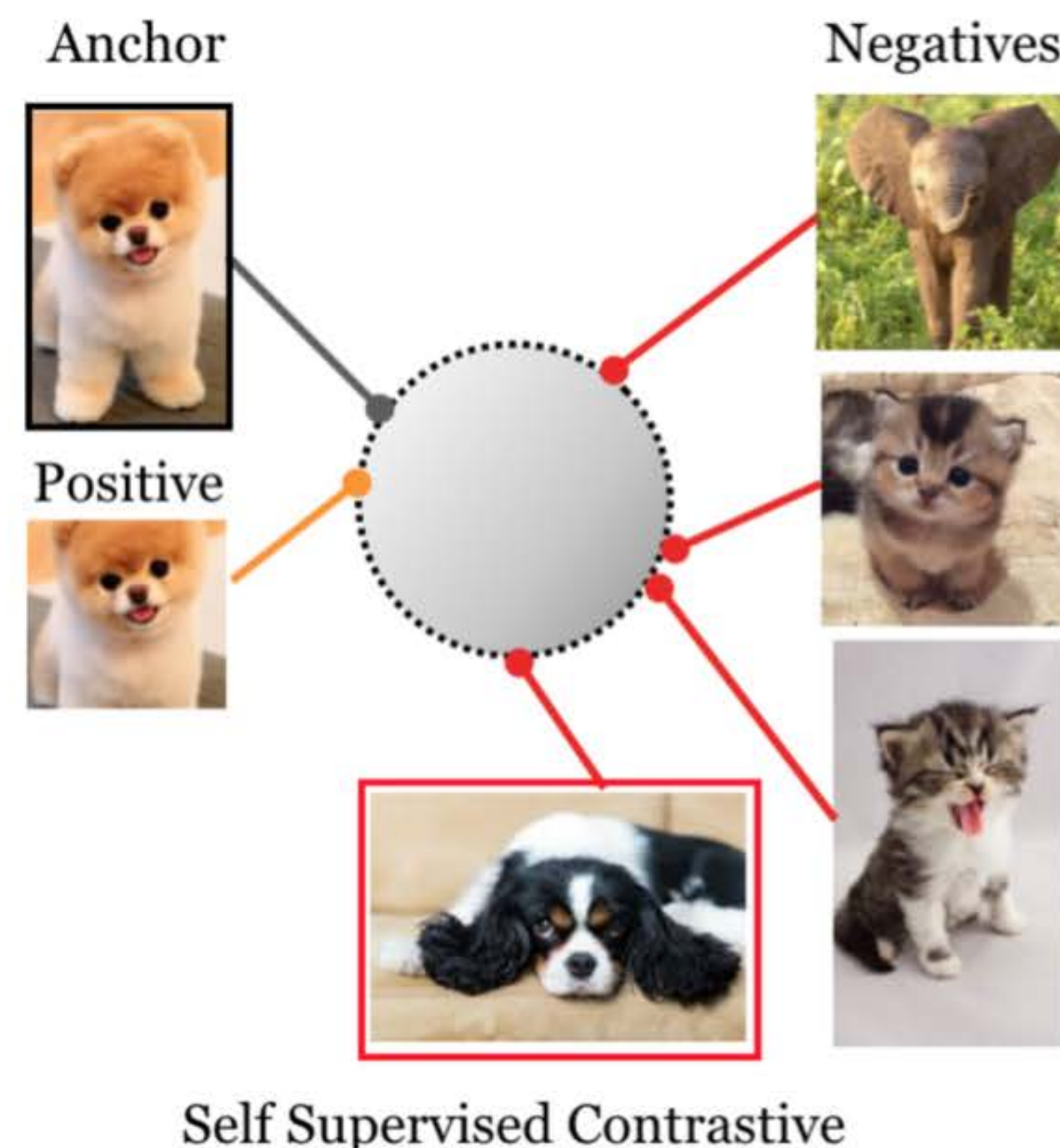
- We are pushing apart negative pairs. Negative pairs are random pairs from the data.



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Supervised or semi-supervised contrastive learning

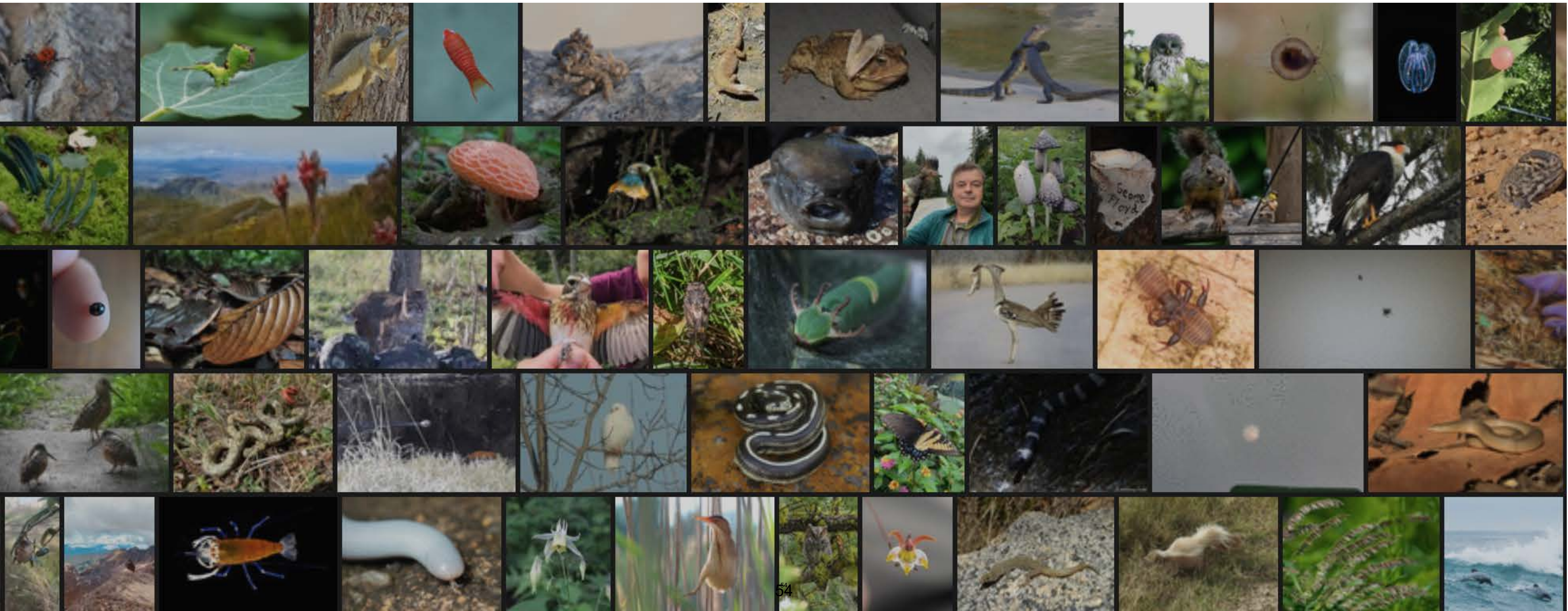
- Contrastive learning provides more geometric and robustness feedback than cross-entropy loss
- Idea: in addition to data augmentation, use images from same class as positive pairs (multiple positive pairs)

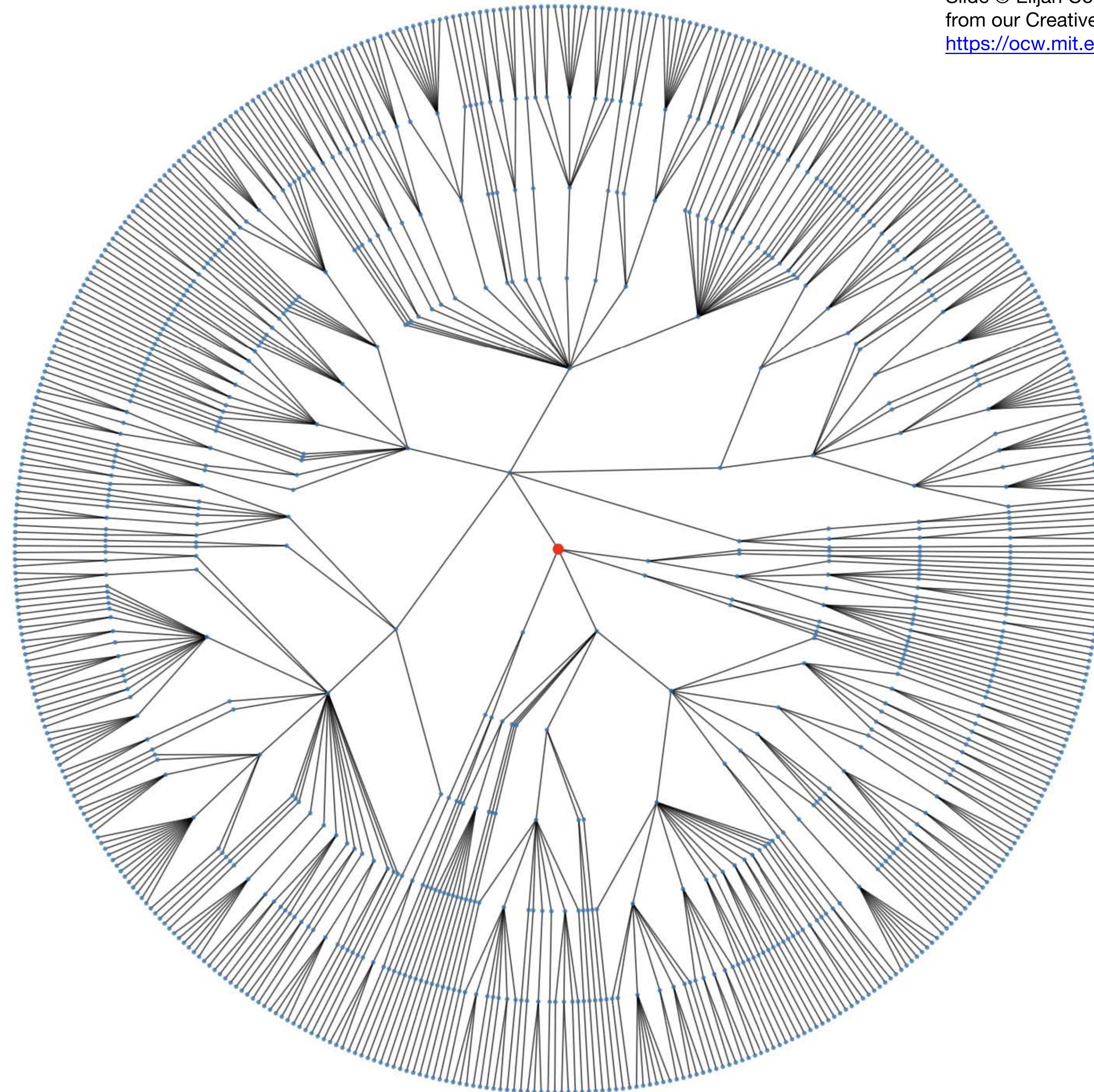


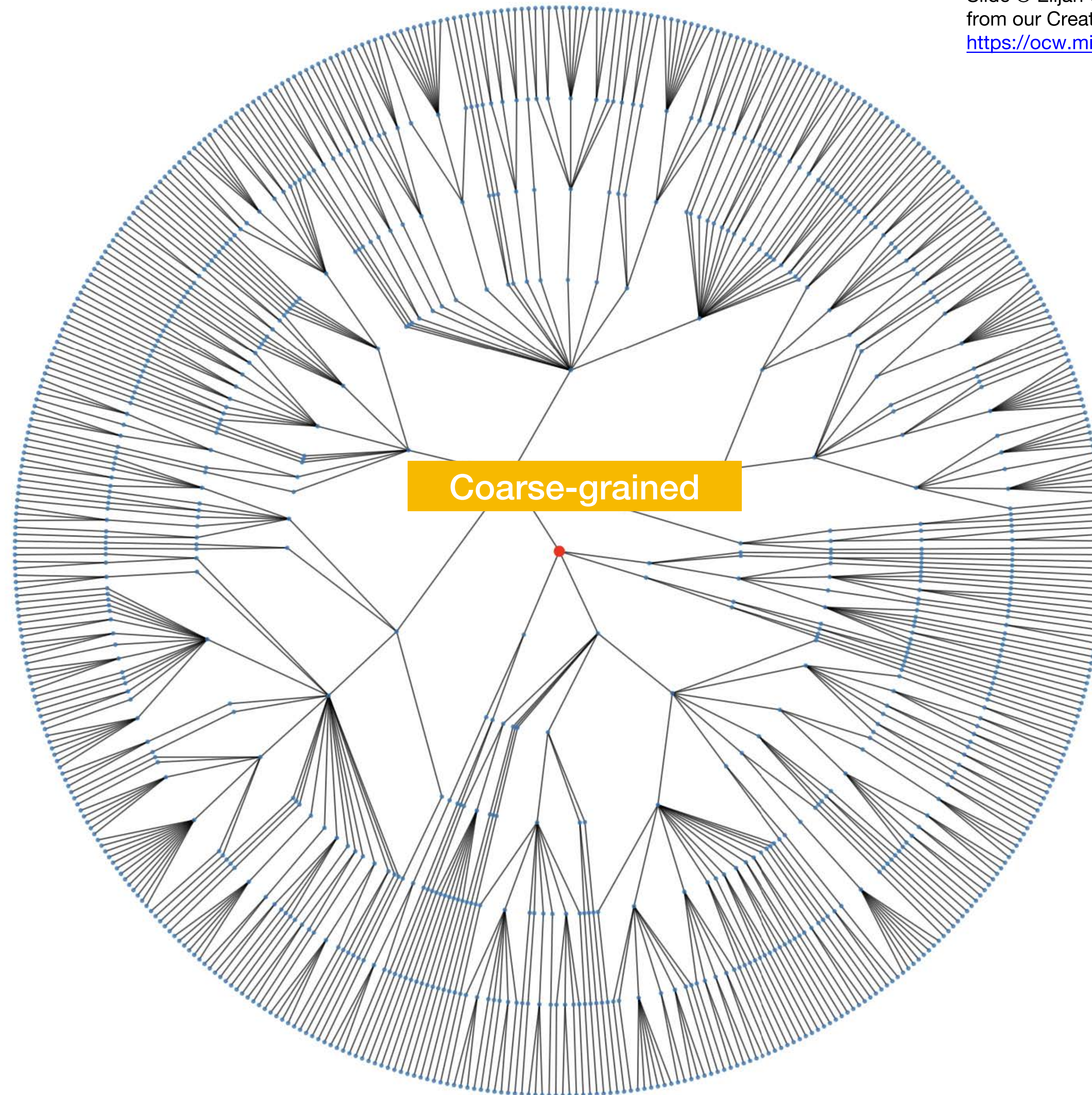
Case study: iNaturalist 2021

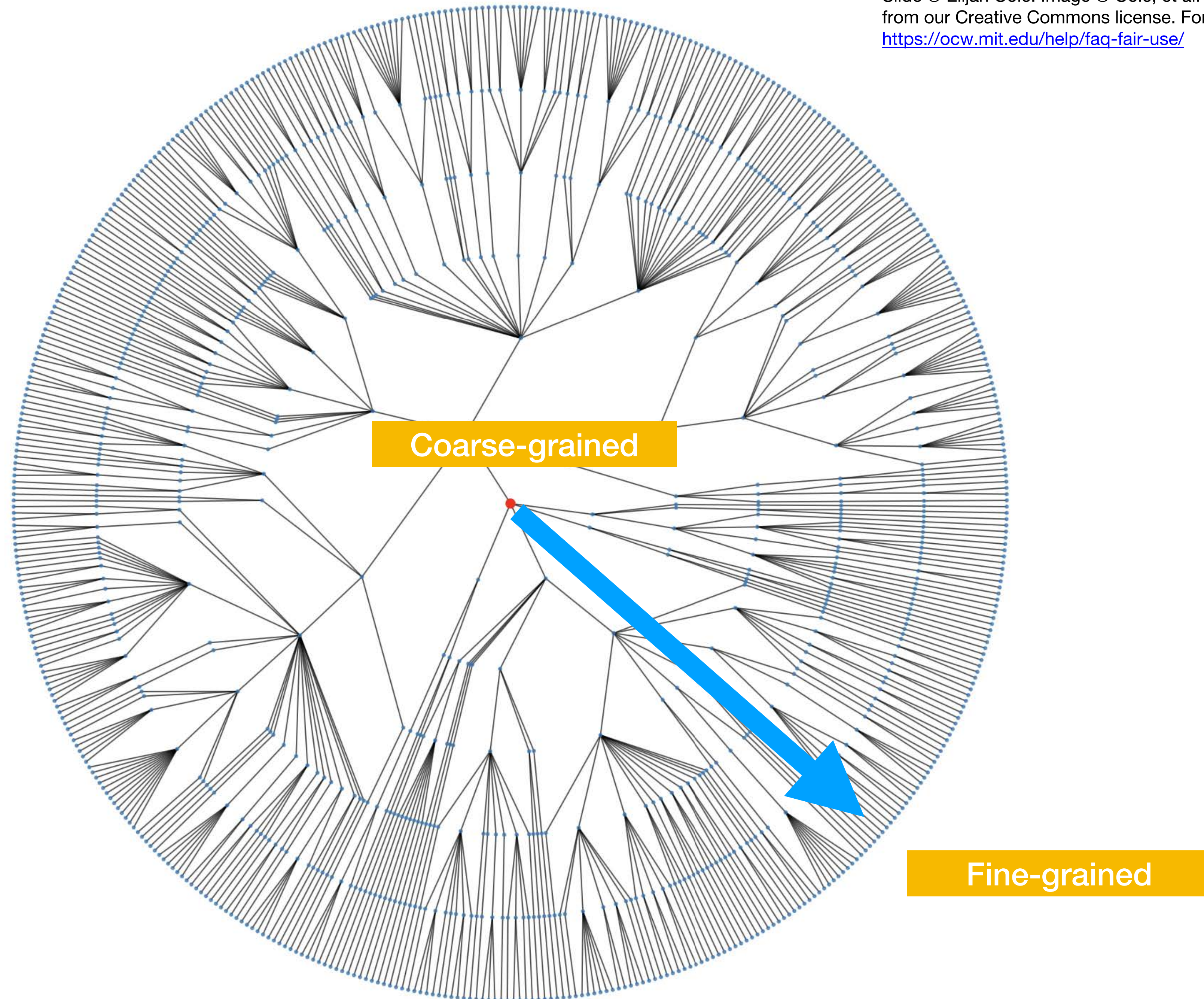
- 10,000 Species
- 2.7M Training Images
- 50k Validation Images
- 500k Test Images

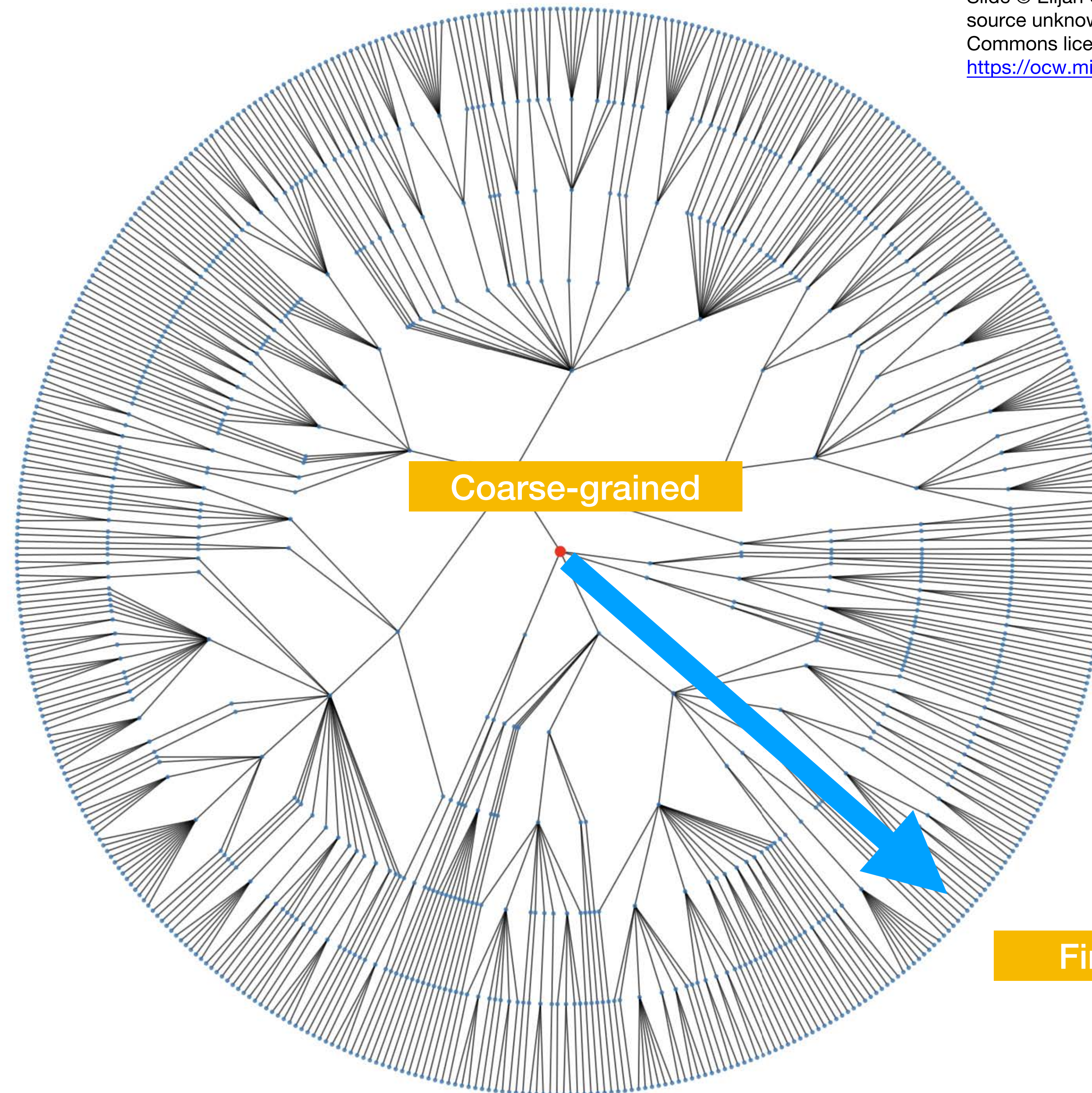
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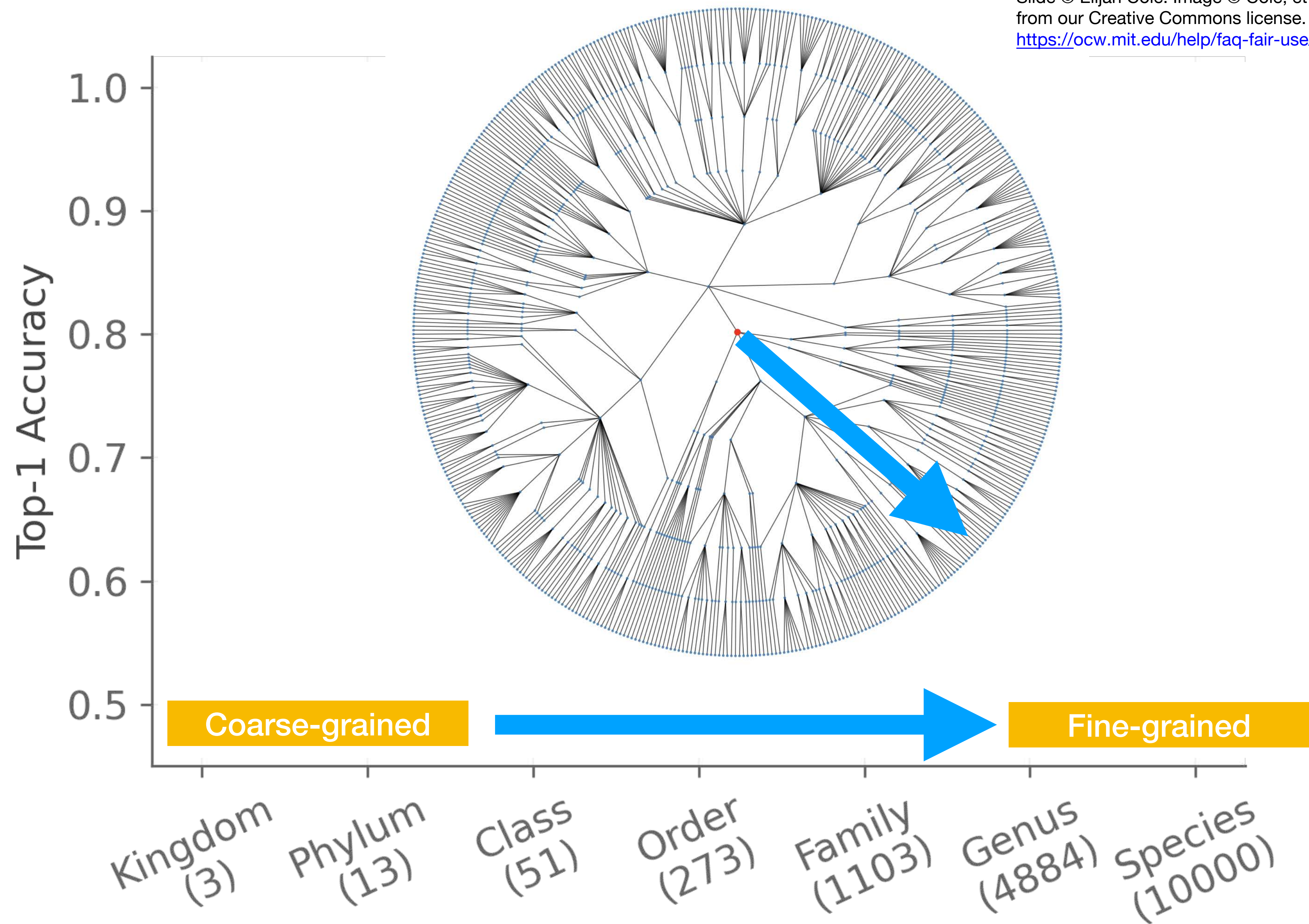


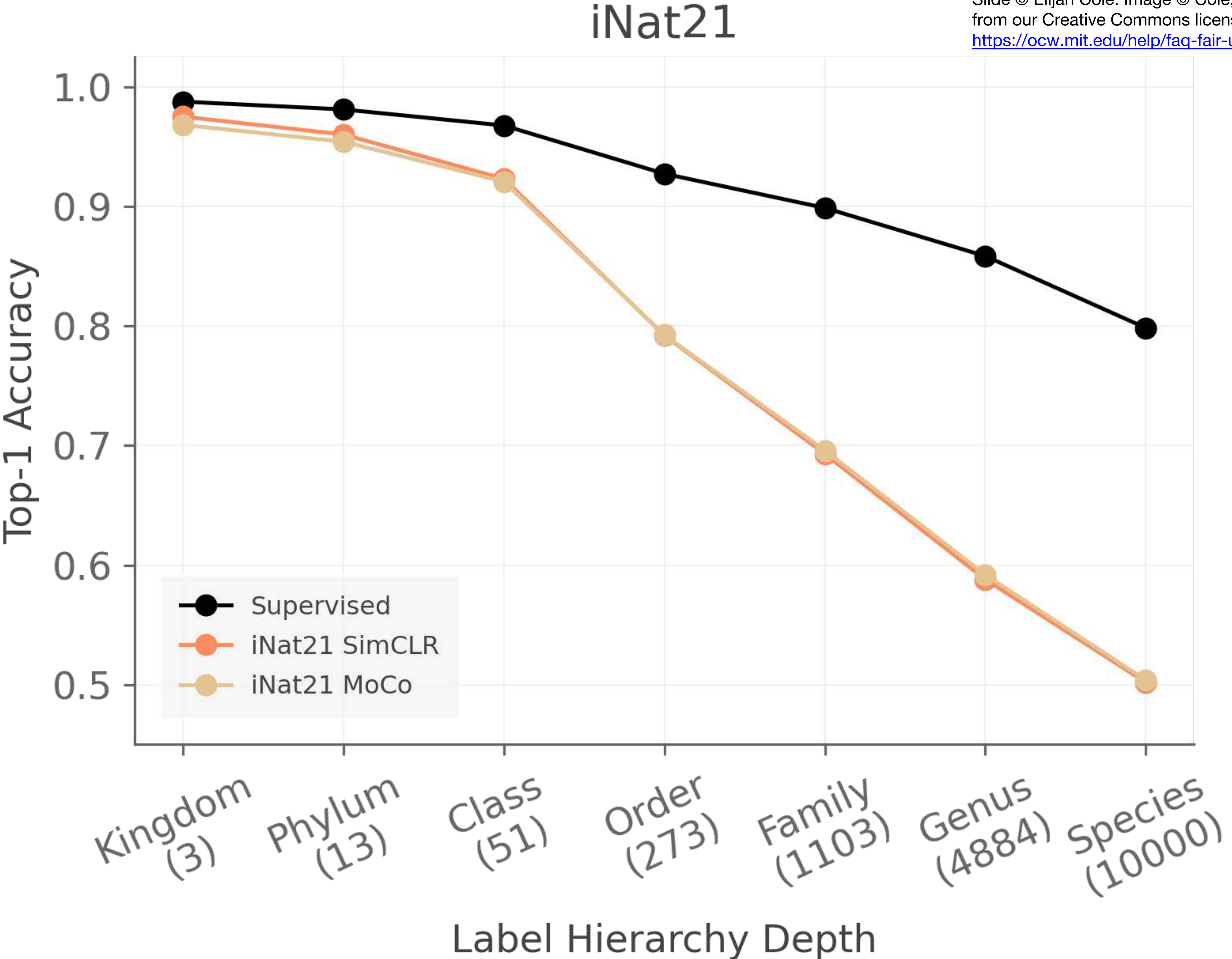
S. umbilicata

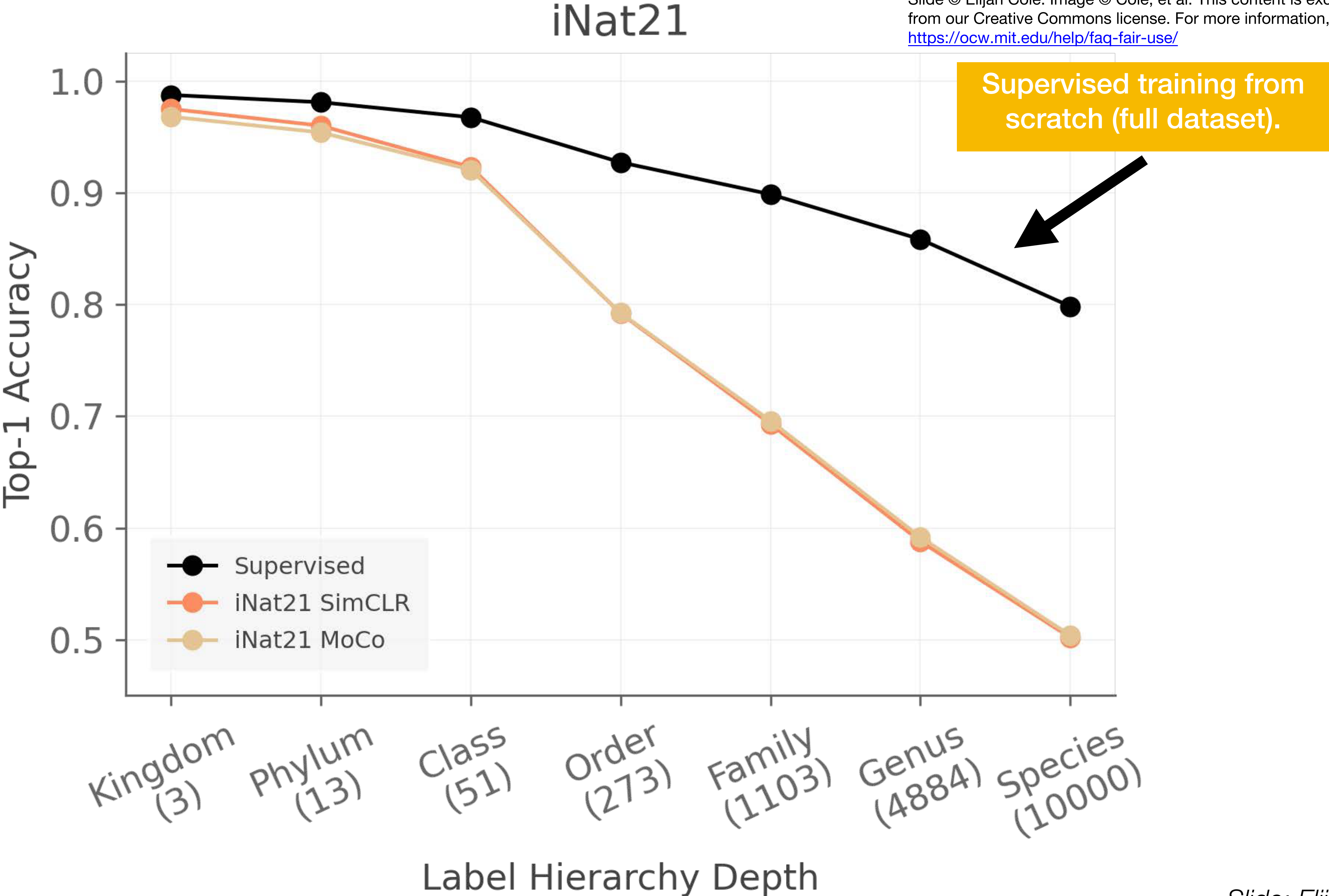


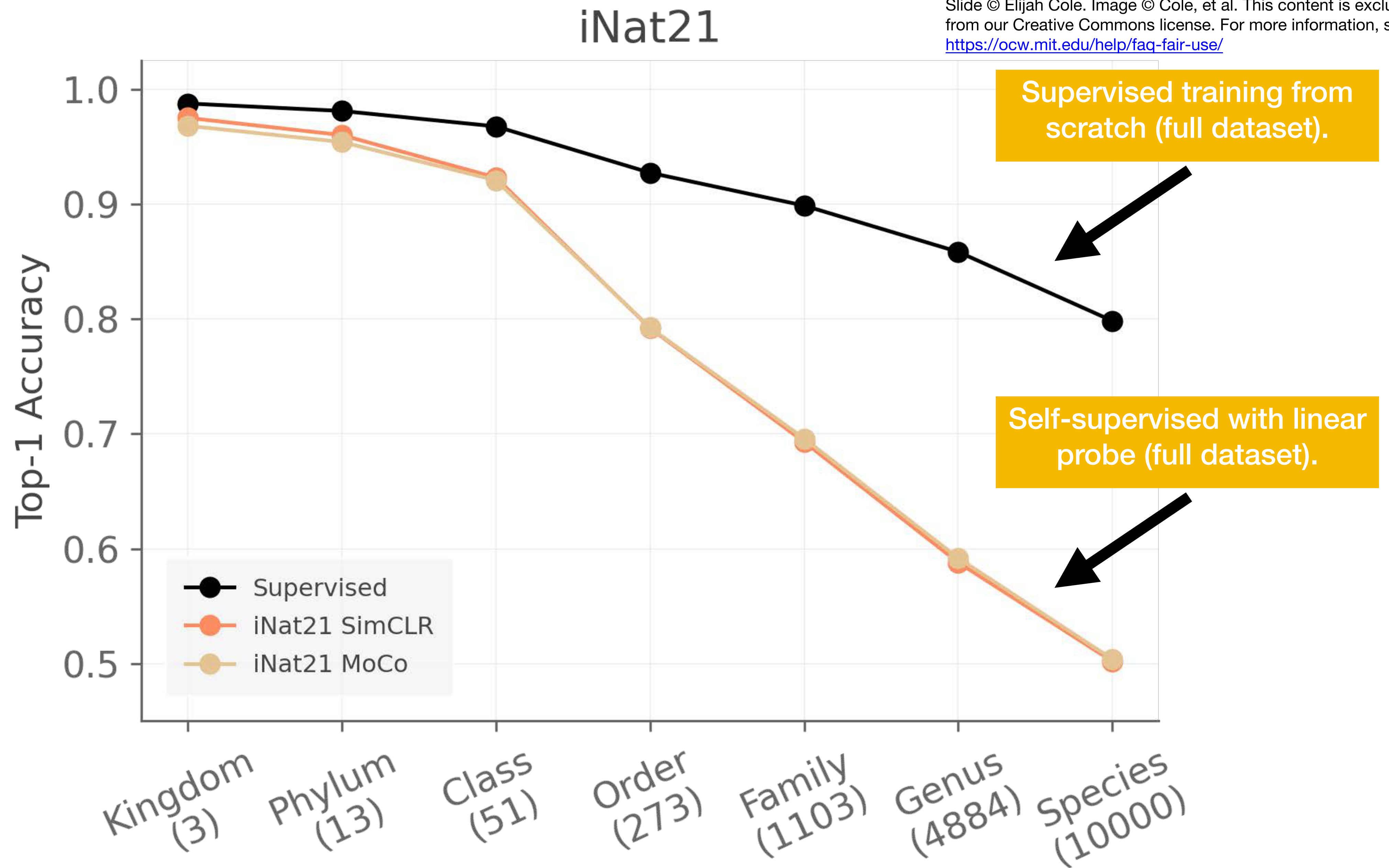
S. ornata

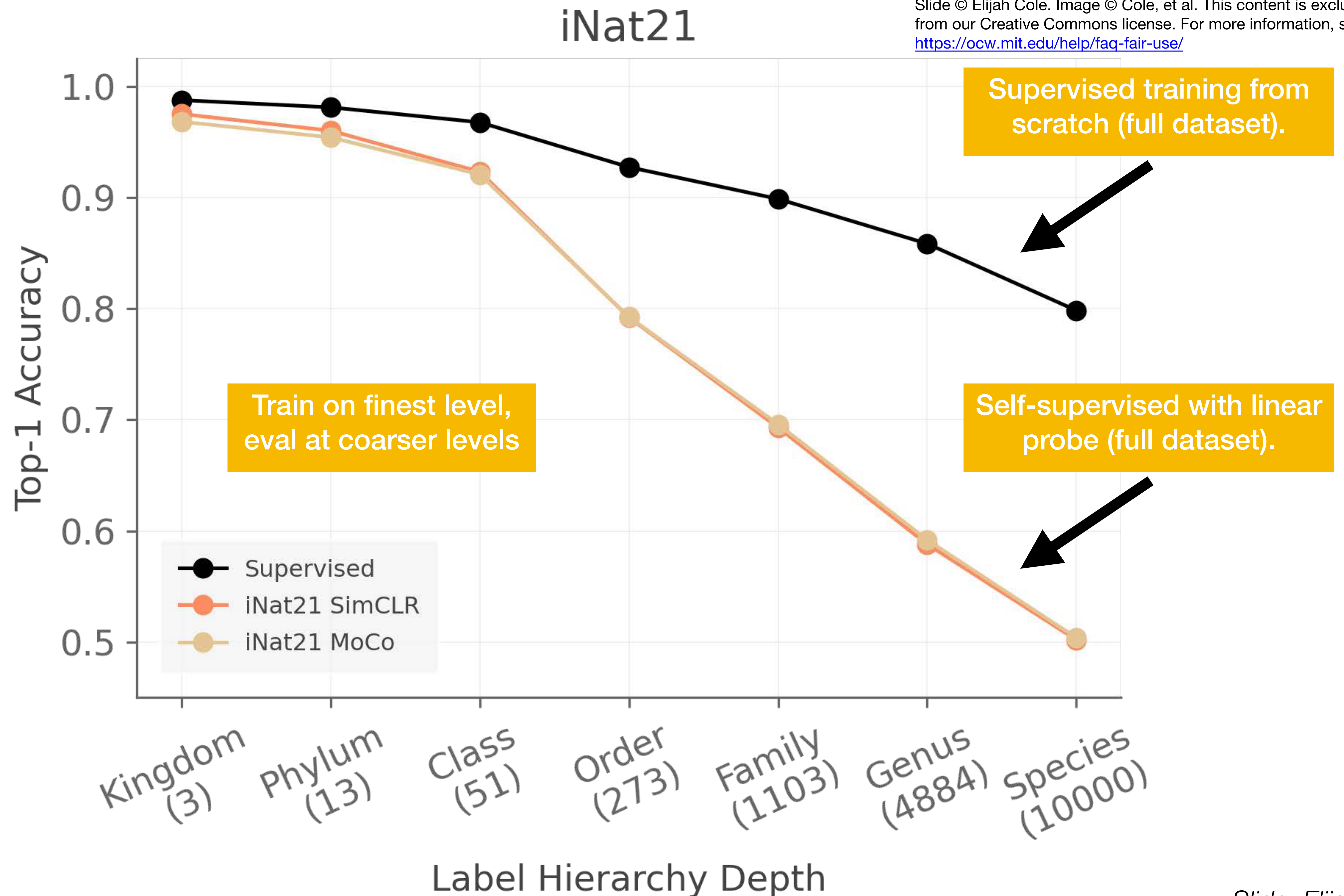


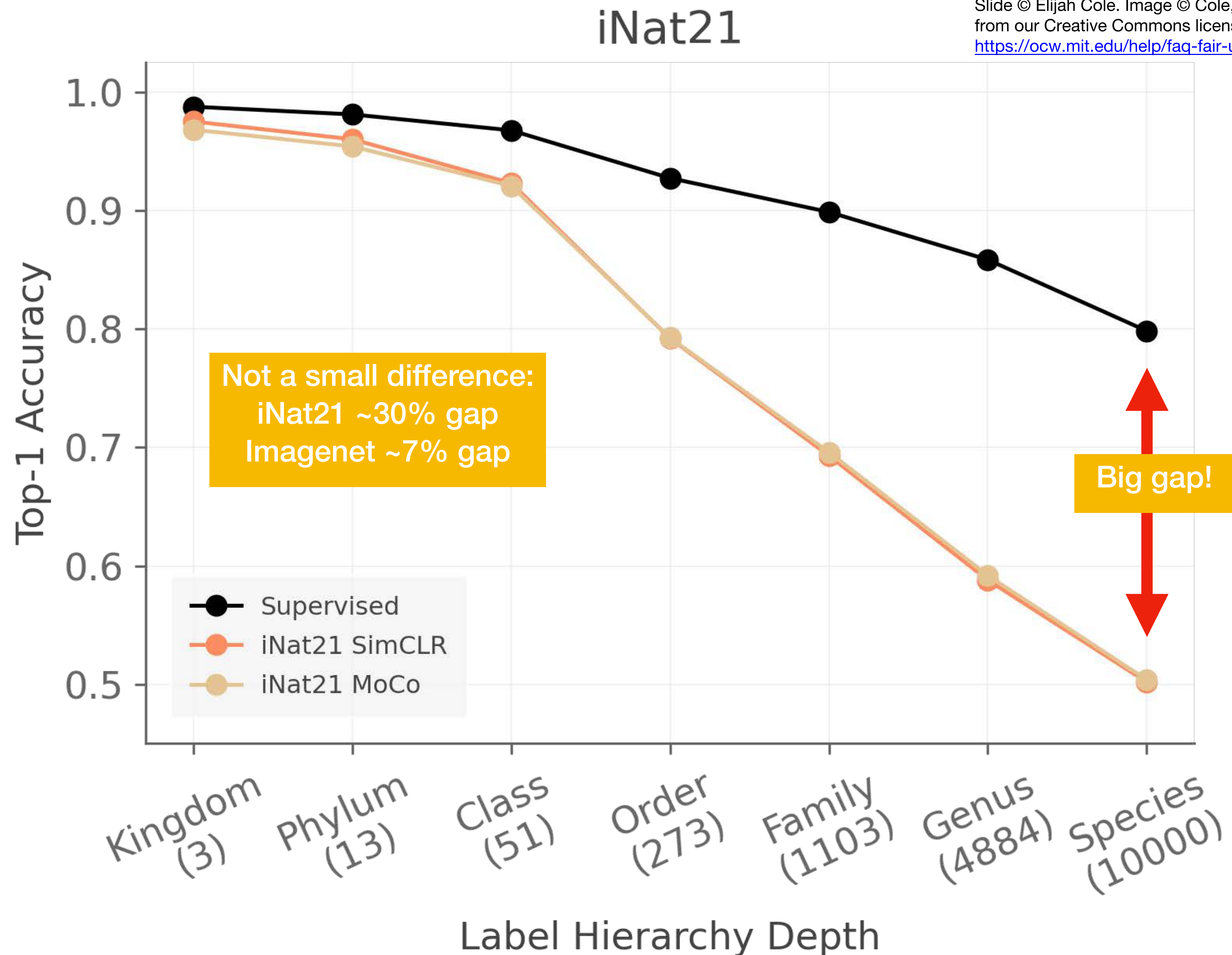


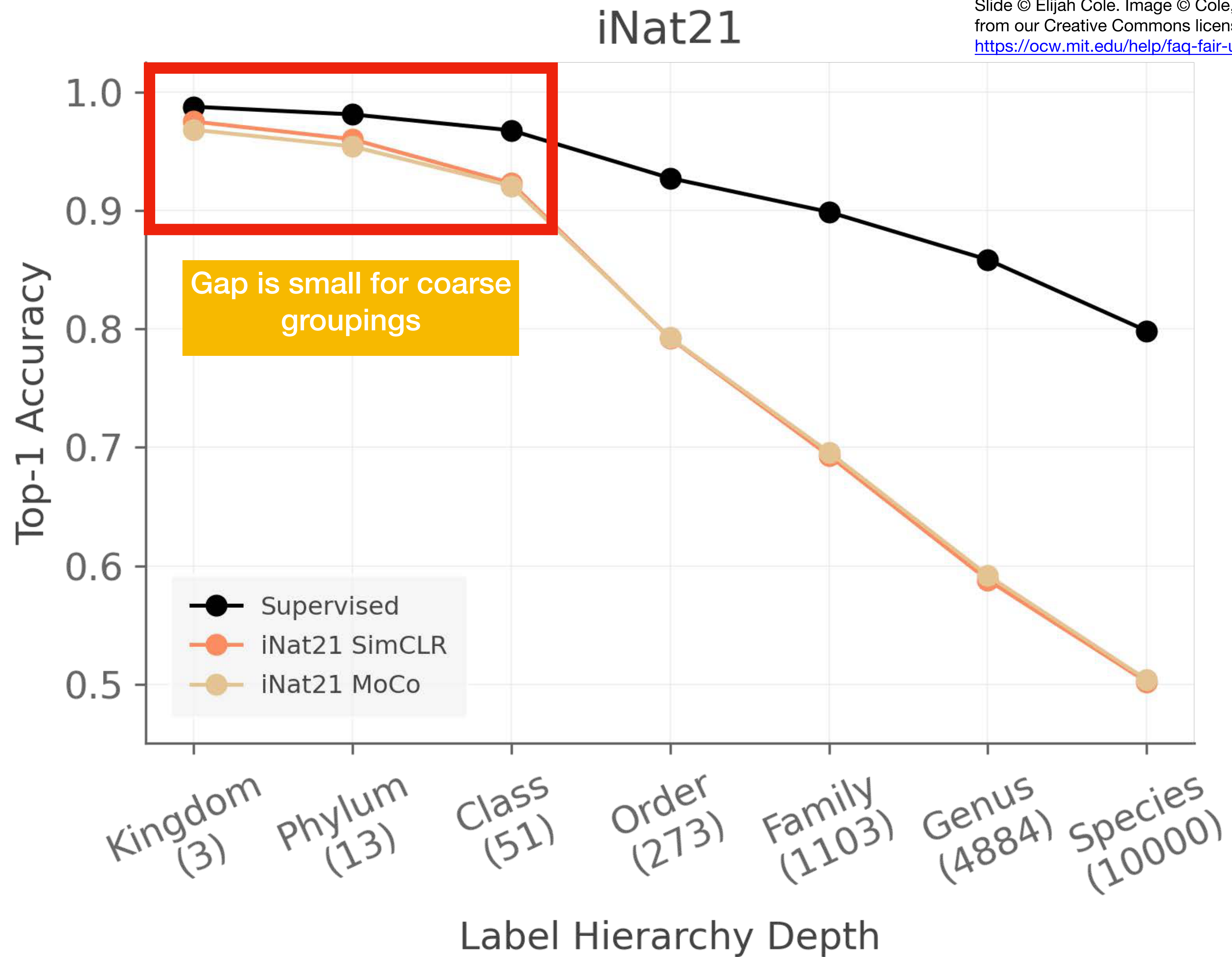


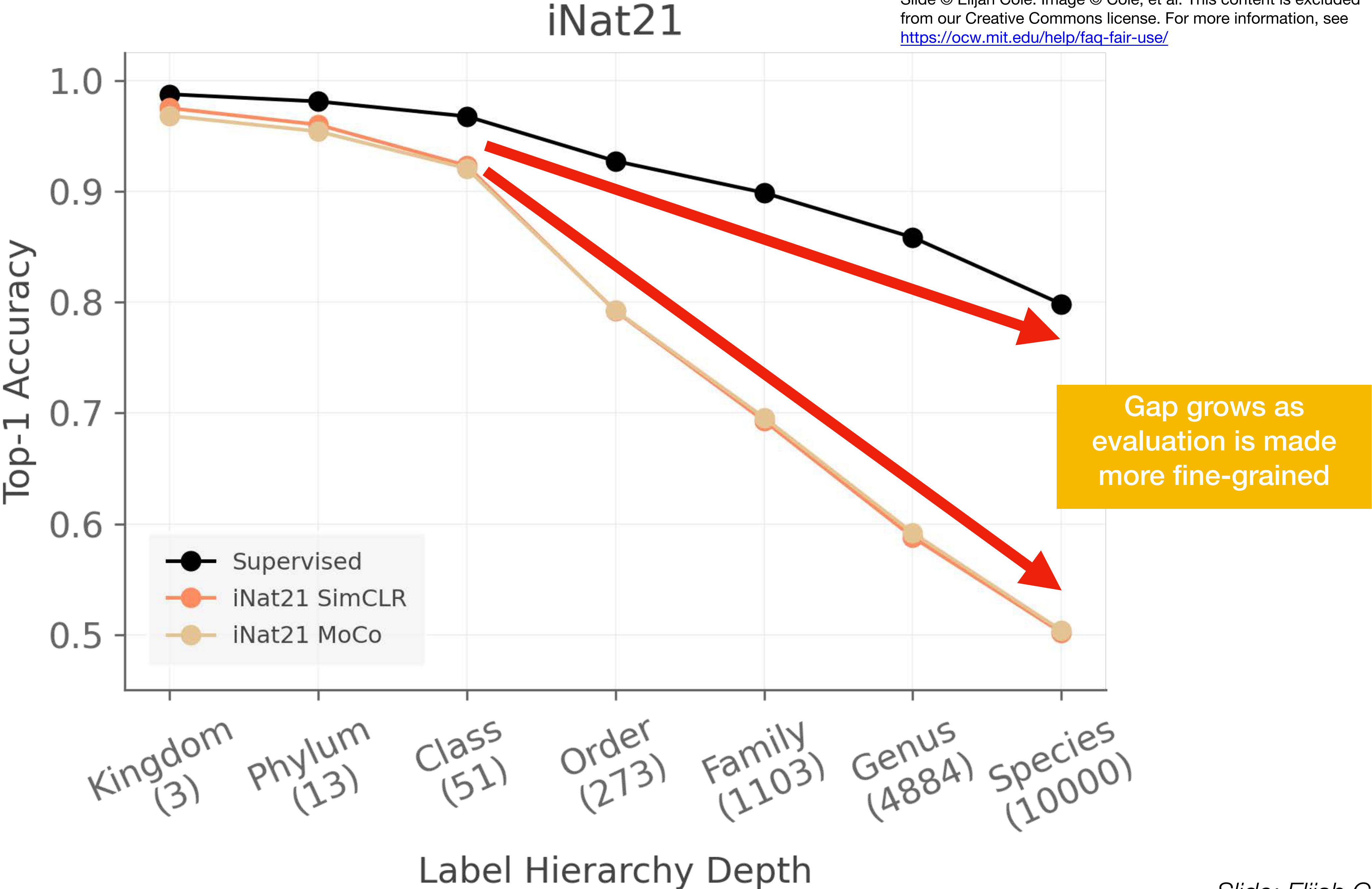


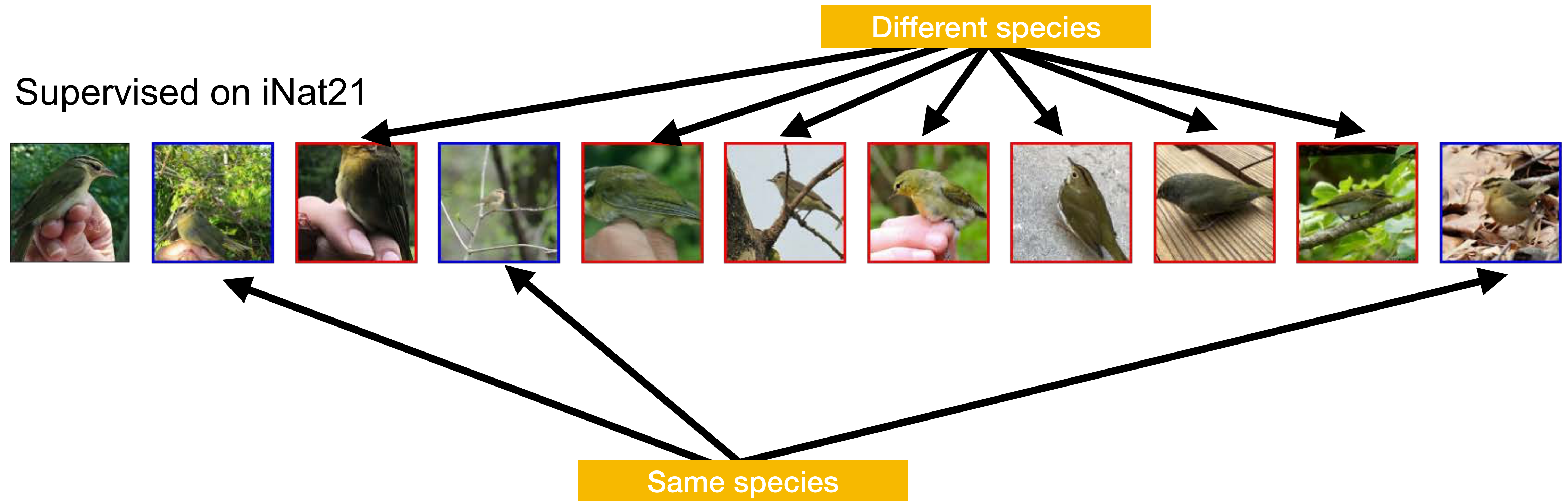












On ImageNet, contrastive SSL matches supervised.

On iNat21, contrastive SSL lags far behind

Supervised on iNat21



SimCLR on iNat21



Summary

- Good representations capture relevant similarity/dissimilarity information
- well-clustered, compact and separated/spread out classes:
 - preserves relevant information
 - teaches relevant invariances (“forget” irrelevant information)
- supervised or self-supervised

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6.7960 Deep Learning

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